

TRIGONOMETRY UPTO TRANSFORMATIONS

WORKEDOUT EXAMPLES:

W.E-1: If $\tan\left(\frac{\pi}{4} + \frac{y}{2}\right) = \tan^3\left(\frac{\pi}{4} + \frac{x}{2}\right)$ then

$$\frac{3 + \sin^2 x}{1 + 3 \sin^2 x} =$$

Ans: $\frac{\sin y}{\sin x}$

Sol:
$$\frac{1 + \tan \frac{y}{2}}{1 - \tan \frac{y}{2}} = \left(\frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right)^3$$

Squaring on both sides

$$\frac{1 + \sin y}{1 - \sin y} = \left(\frac{1 + \sin x}{1 - \sin x} \right)^3$$

Apply componendo and dividendo rule

$$\frac{2 \sin y}{2} = \frac{(3 + \sin^2 x) \sin x}{1 + 3 \sin^2 x}$$

$$\frac{3 + \sin^2 x}{1 + 3 \sin^2 x} = \frac{\sin y}{\sin x}$$

W.E-2: The minimum value of

$$\frac{\sec^4 \alpha}{\tan^2 \beta} + \frac{\sec^4 \beta}{\tan^2 \alpha}, \text{ for all } \alpha, \beta \neq \frac{k\pi}{2},$$

Ans: $k \in \mathbb{Z}$ is 8

Sol: Let $\tan^2 \alpha = a, \tan^2 \beta = b$
We have to determine the minimum value of

$$\frac{(a+1)^2}{b} + \frac{(b+1)^2}{a} \text{ where } a, b \geq 0$$

$$\frac{(a+1)^2}{b} + \frac{(b+1)^2}{a} = \left(\frac{a^2}{b} + \frac{1}{b} + \frac{b^2}{a} + \frac{1}{a} \right) + 2 \left(\frac{a}{b} + \frac{b}{a} \right)$$

$$\geq 4 \left(\frac{a^2}{b} \cdot \frac{1}{b} \cdot \frac{b^2}{a} \cdot \frac{1}{a} \right)^{\frac{1}{4}} + 4 \left(\frac{a}{b} \cdot \frac{b}{a} \right)^{\frac{1}{2}}$$

$$\geq 4 + 4 \geq 8$$

W.E-3: If $\frac{\tan(A-B)}{\tan A} + \frac{\sin^2 C}{\sin^2 A} = 1$

then $\tan A, \tan C, \tan B$ are in

Ans: G.P

Sol:
$$\frac{\sin^2 C}{\sin^2 A} = 1 - \frac{\tan(A-B)}{\tan A}$$

$$\frac{\sin^2 C}{\sin^2 A} = \frac{\tan B(1 + \tan^2 A)}{\tan A(1 + \tan A \tan B)} = \left(\frac{\tan B \sec^2 A}{\tan A(1 + \tan A \tan B)} \right)$$

$$\sin^2 C(1 + \tan A \tan B) = \tan A \tan B$$

$$\sin^2 C = \tan A \tan B(1 - \sin^2 C)$$

$$\tan^2 C = \tan A \tan B$$

W.E-4: If $\cot^2 x = \cot(x-y)\cot(x-z)$ then

$\cot(2x)$ is equal to (where $x \neq \pm \frac{\pi}{4}$)

Ans: $\frac{1}{2}(\cot y + \cot z)$

Sol:
$$\cot^2 x = \left(\frac{\cot x \cot y + 1}{\cot y - \cot x} \right) \left(\frac{\cot x \cot z + 1}{\cot z - \cot x} \right)$$

$$\Rightarrow \cot^3 x(\cot y + \cot z) + \cot x(\cot y + \cot z) + 1 - \cot^4 x = 0$$

$$\cot x(1 + \cot^2 x)(\cot y + \cot z) + (1 + \cot^2 z)(1 - \cot^2 x) = 0$$

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$$\frac{\cot^2 x - 1}{2 \cot x} = \frac{1}{2}(\cot y + \cot z)$$

$$\cot 2x = \frac{1}{2}(\cot y + \cot z)$$

W.E-5: If

$$(\tan x - \tan y)^2, (\tan y - \tan z)^2 \text{ and } (\tan z - \tan x)^2$$

are in A.P then $\tan x - \tan y$, $\tan y - \tan z$, $\tan z - \tan x$ are in

Ans: H.P

Sol: Let

$$a = \tan x - \tan y, b = \tan y - \tan z, c = \tan z - \tan x$$

$$\text{then } a + b + c = 0$$

$$b = -(a + c)$$

$$b^2 = a^2 + c^2 + 2ac$$

$$\text{given } 2b^2 = a^2 + c^2, b^2 = -2ac$$

$$-b = \frac{2ac}{b} = \frac{2ac}{-(a+c)}, b = \frac{2ac}{a+c}$$

$\tan x - \tan y$, $\tan y - \tan z$, $\tan z - \tan x$ are in H.P.

W.E-6: If $\tan \alpha$, $\tan \beta$, $\tan \gamma$ are the roots of the

equation $x^3 - px^2 - r = 0$ then the value of

$$(1 + \tan^2 \alpha)(1 + \tan^2 \beta)(1 + \tan^2 \gamma) =$$

$$\text{Ans: } 1 + (p - r)^2$$

$$\text{Sol: } \sum \tan \alpha = p, \sum \tan \alpha \tan \beta = 0, \tan \alpha \tan \beta \tan \gamma = r$$

Required value =

$$1 + \sum \tan^2 \alpha + \sum \tan^2 \alpha \tan^2 \beta + \tan^2 \alpha \tan^2 \beta \tan^2 \gamma$$

$$= 1 + (\sum \tan \alpha)^2 - 2 \sum \tan \alpha \tan \beta + (\sum \tan \alpha \tan \beta)^2 -$$

$$2 \tan \alpha \tan \beta \tan \gamma \sum \tan \alpha + (\tan \alpha \tan \beta \tan \gamma)^2$$

$$= 1 + p^2 - 2pr + r^2 = 1 + (p - r)^2$$

W.E-7: If $\tan \theta = n \tan \phi$ then the maximum value of

$\tan^2 (\theta - \phi)$ is equal to

$$\text{Ans: } \frac{(n-1)^2}{4n}$$

$$\text{Sol: } \tan (\theta - \phi) = \frac{\tan \theta - \tan \phi}{1 + \tan \theta \tan \phi} = \frac{(n-1) \tan \phi}{1 + n \tan^2 \phi}$$

$$\tan (\theta - \phi) = \frac{n-1}{\cot \phi + n \tan \phi}$$

$$\cot \phi + n \tan \phi \geq 2\sqrt{n}$$

$$\tan^2 (\theta - \phi) = \frac{(n-1)^2}{(\cot \phi + n \tan \phi)^2}$$

$$\text{maximum value of } \tan^2 (\theta - \phi) = \frac{(n-1)^2}{4n}$$

W.E-8: If $\left| \cos \theta \left(\sin \theta + \sqrt{\sin^2 \theta + \sin^2 \alpha} \right) \right| \leq k$

then the value of k is

$$\text{Ans: } \sqrt{1 + \sin^2 \alpha}$$

$$\text{Sol: } x = \cos \theta \sin \theta + \cos \theta \sqrt{\sin^2 \theta + \sin^2 \alpha}$$

Squaring on both sides

$$x^2 + \cos^2 \theta \sin^2 \theta - 2x \cos \theta \sin \theta$$

$$= \cos^2 \theta \sin^2 \theta + \cos^2 \theta \sin^2 \alpha$$

$$x^2 - 2x \cos \theta \sin \theta - \cos^2 \theta \sin^2 \alpha = 0$$

dividing by $\cos^2 \theta$

$$x^2 (1 + \tan^2 \theta) - 2x \tan \theta - \sin^2 \alpha = 0$$

$$x^2 \tan^2 \theta - 2x \tan \theta + (x^2 - \sin^2 \alpha) = 0$$

Since $\tan \theta$ is real $\Delta \geq 0$

$$4x^2 - 4x^2 (x^2 - \sin^2 \alpha) \geq 0$$

$$4x^2 (1 - x^2 + \sin^2 \alpha) \geq 0$$

$$x^2 \leq 1 + \sin^2 \alpha$$

$$|x| \leq \sqrt{1 + \sin^2 \alpha}$$

W.E-9: The value of

$$2 \sin 2^0 + 4 \sin 4^0 + 6 \sin 6^0 + \dots + 178 \sin 178^0 =$$

$$\text{Ans: } 90 \cot 1^0$$

Sol:

$$x = 2 \sin 2^0 + 4 \sin 4^0 + 6 \sin 6^0 + \dots + 178 \sin 178^0$$

$$x = (2+178) \sin 2^0 + (4+176) \sin 4^0 + \dots + 90 \sin 90^0$$

$$= 180 [\sin 2^0 + \sin 4^0 + \dots + \sin 88^0] + 90$$

$$\begin{aligned}
 x \sin 1^\circ &= 90 [2 \sin 2^\circ \sin 1^\circ + 2 \sin 4^\circ \sin 1^\circ + \dots + 2 \sin 88^\circ \sin 1^\circ] + 90 \sin 1^\circ \\
 &= 90 [\cos 1^\circ - \cos 3^\circ + \cos 3^\circ - \cos 5^\circ + \dots + \cos 87^\circ - \cos 89^\circ] + 90 \sin 1^\circ \\
 &= 90 [\cos 1^\circ - \cos 89^\circ + \sin 1^\circ] = 90 [\cos 1^\circ - \sin 1^\circ + \sin 1^\circ] \\
 x &= 90 \cot 1^\circ
 \end{aligned}$$

W.E-10: The value of

$$\frac{2 \sec \theta + 3 \tan \theta + 5 \sin \theta - 7 \cos \theta + 5}{2 \tan \theta + 3 \sec \theta + 5 \cos \theta + 7 \sin \theta + 8} =$$

Ans: $\tan\left(\frac{\theta}{2}\right)$

Sol: $C = \cos \theta, S = \sin \theta$

$$E = \frac{2 + 3S + 5CS - 7C^2 + 5C}{2S + 3 + 5C^2 + 7CS + 8C}$$

$$= \frac{S(5C + 3) - (7C + 2)(C - 1)}{S(7C + 2) + (C + 1)(5C + 3)}$$

$$S^2 = 1 - C^2 = (1 - C)(1 + C)$$

$$\frac{S}{1 - C} = \frac{1 + C}{S} = \lambda \text{ say}$$

$$= \frac{(1 - C)\lambda(5C + 3) + (1 - C)(7C + 2)}{S(7C + 2) + \lambda S(5C + 3)}$$

$$= \frac{1 - C}{S} = \tan \frac{\theta}{2}$$

W.E-11: Let $f(\theta) = \frac{1}{1 + (\cot \theta)^x}$ and $S = \sum_{\theta=1^\circ}^{89^\circ} f(\theta)$

Then the value of $\sqrt{2S - 8} =$

Ans: 9

Sol: $f(\theta) = \frac{(\sin \theta)^x}{(\sin \theta)^x + (\cos \theta)^x}$

$$f\left(\frac{\pi}{2} - \theta\right) + f(\theta) = 1$$

$$S = f(1^\circ) + f(2^\circ) + \dots + f(89^\circ)$$

$$= (f(1^\circ) + f(89^\circ)) + (f(2^\circ) + f(88^\circ)) + \dots + (f(44^\circ) + f(46^\circ)) + f(45^\circ)$$

$$= (1 + 1 + \dots + 1 \text{ 44 times}) + \frac{1}{2} = \frac{89}{2}$$

$$\sqrt{2S - 8} = 9$$

W.E-12: Let $0 \leq a, b, c, d \leq \pi$ where b, c are not complementary such that

$$\begin{aligned}
 2 \cos a + 6 \cos b + 7 \cos c + 9 \cos d &= 0 \text{ and} \\
 2 \sin a - 6 \sin b + 7 \sin c - 9 \sin d &= 0
 \end{aligned}$$

Then the value of $\frac{\cos(a + d)}{\cos(b + c)} =$

Ans: $\frac{7}{3}$

Sol: $(2 \cos a + 9 \cos d)^2 = (6 \cos b + 7 \cos c)^2$

$$\begin{aligned}
 4 \cos^2 a + 81 \cos^2 d + 36 \cos a \cos d \\
 = 36 \cos^2 b + 49 \cos^2 c + 84 \cos b \cos c - (1)
 \end{aligned}$$

$$(2 \sin a - 9 \sin d)^2 = (6 \sin b - 7 \sin c)^2$$

$$\begin{aligned}
 4 \sin^2 a + 81 \sin^2 d - 36 \sin a \sin d \\
 = 36 \sin^2 b + 49 \sin^2 c - 84 \sin b \sin c - (2)
 \end{aligned}$$

Adding (1) and (2)

$$4 + 81 + 36 \cos(a + d) = 36 + 49 + 84 \cos(b + c)$$

$$\frac{\cos(a + d)}{\cos(b + c)} = \frac{84}{36} = \frac{7}{3}$$

W.E-13: If x, y, z, t are real numbers such that

$$\begin{aligned}
 x^2 + y^2 = 9, z^2 + t^2 = 4 \text{ and } xt - yz = 6 \\
 \text{then the greatest value of } xz \text{ is}
 \end{aligned}$$

Ans: 3

Sol: Let $x = 3 \cos \theta$ $y = 3 \sin \theta$

$$z = 2 \cos \phi \quad t = 2 \sin \phi$$

$$\Rightarrow 6 \cos \theta \sin \phi - 6 \sin \theta \cos \phi = 6$$

$$\sin(\phi - \theta) = 1 \quad \phi - \theta = 90^\circ \Rightarrow \phi = 90^\circ + \theta$$

$$xz = -6 \sin \theta \cos \theta = -3 \sin 2\theta \leq 3$$

W.E-14: If $x, y \in \left(0, \frac{\pi}{2}\right)$ and $2 \sin x = \sin y, 2 \cos x = 3 \cos y$

then $[\tan(x + y)]$ (where $[\cdot]$ denote the G.I.F) is

Ans: 3

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$$\text{Sol: } \frac{\sin x}{\sin y} = \frac{1}{2}, \quad \frac{\cos x}{\cos y} = \frac{3}{2}, \quad \frac{\tan x}{\tan y} = \frac{1}{3}$$

$$\tan(x+y) = \frac{4 \tan x}{1 - 3 \tan^2 x}$$

$$\sin y = 2 \sin x \quad \cos y = \frac{2}{3} \cos x$$

$$\sin^2 y + \cos^2 y = 4 \sin^2 x + 4 \frac{\cos^2 x}{9} = 1,$$

$$\tan^2 x = \frac{5}{27}$$

$$\tan^2(x+y) = 15, \quad \tan(x+y) = \sqrt{15}, \text{ and } [\tan(x+y)] = 3$$

W.E-15: If $\sec A \tan B + \tan A \sec B = 91$ then the value of $(\sec A \sec B + \tan A \tan B)^2 - 91^2$ is equal to

Ans: 1

Sol:

$$S = (\sec A \sec B + \tan A \tan B)^2 - (\sec A \tan B + \tan A \sec B)^2$$

$$= \left(\frac{1 + \sin A \sin B}{\cos A \cos B} \right)^2 - \left(\frac{\sin A + \sin B}{\cos A \cos B} \right)^2$$

$$= \frac{1 + \sin^2 A \sin^2 B - \sin^2 A - \sin^2 B}{\cos^2 A \cos^2 B} = \frac{\cos^2 A \cos^2 B}{\cos^2 A \cos^2 B} = 1$$

$$(\sec A \sec B + \tan A \tan B)^2 - 91^2 = 1$$

W.E-16: If

$$\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -\frac{3}{2} \text{ then}$$

$$\text{A) } \sum \cos \alpha = 0 \quad \text{B) } \sum \sin \alpha = 0$$

$$\text{C) } \sum (\cos \alpha + \sin \alpha) = 0 \quad \text{D) } \sum \cos \alpha \sin \alpha = 0$$

Ans: A,B,C

Sol:

$$2 \left(\cos \alpha \cos \beta + \sin \alpha \sin \beta + \cos \beta \cos \gamma + \sin \beta \sin \gamma + \cos \gamma \cos \alpha + \sin \gamma \sin \alpha \right)$$

$$+ \left(\cos^2 \alpha + \sin^2 \alpha + \cos^2 \beta + \sin^2 \beta + \cos^2 \gamma + \sin^2 \gamma \right) = 0$$

$$(\cos \alpha + \cos \beta + \cos \gamma)^2 + (\sin \alpha + \sin \beta + \sin \gamma)^2 = 0$$

$$\sum \cos \alpha = 0 \text{ and } \sum \sin \alpha = 0$$

W.E-17: If

$$\tan \theta + \tan \phi = a \text{ and } \cot \theta + \cot \phi = b, \text{ and } \theta - \phi = \alpha \neq 0$$

then

$$\text{A) } ab > 4$$

$$\text{B) } ab = 4$$

$$\text{C) } \tan^2 \alpha = \frac{ab(ab-4)}{(a+b)^2}$$

$$\text{D) } \cot^2 \alpha = \frac{ab(ab+4)}{(a-b)^2}$$

Ans: A,C

$$\text{Sol: } \frac{a}{b} = \tan \theta \tan \phi$$

$$\begin{aligned} \tan^2 \alpha &= \tan^2(\theta - \phi) = \left(\frac{\tan \theta - \tan \phi}{1 + \tan \theta \tan \phi} \right)^2 \\ &= \frac{a^2 - 4 \left(\frac{a}{b} \right)}{\left(1 + \frac{a}{b} \right)^2} = \frac{ab(ab-4)}{(a+b)^2} \end{aligned}$$

$$\tan^2 \alpha > 0, ab - 4 > 0, ab > 4$$

$$\text{W.E-18: } \left(\frac{\sin \theta}{\sin \phi} \right)^2 = \frac{\tan \theta}{\tan \phi} = 3 \text{ then}$$

$$\text{A) } \tan \phi = \frac{1}{\sqrt{3}}$$

$$\text{B) } \tan \phi = \frac{-1}{\sqrt{3}}$$

$$\text{C) } \tan \theta = \sqrt{3}$$

$$\text{D) } \tan \theta = -\sqrt{3}$$

Ans: A,B,C,D

Sol: $\frac{\sin \theta}{\sin \phi} = \frac{\cos \phi}{\cos \theta}$

$\sin 2\theta = \sin \phi$

$\frac{2 \tan \theta}{1 + \tan^2 \theta} = \frac{2 \tan \phi}{1 + \tan^2 \phi}$

$\tan^2 \phi = \frac{1}{3}, \tan \phi = \pm \frac{1}{\sqrt{3}}, \tan \theta = \pm \sqrt{3}$

W.E-19: If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$ then **[IIT 2009]**

A) $\tan^2 x = \frac{2}{3}$ B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

C) $\tan^2 x = \frac{1}{3}$ D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

Ans: A,B

Sol: $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$

$3 \sin^4 x + 2(1 - \sin^2 x)^2 = \frac{6}{5}$

$\Rightarrow 25 \sin^4 x - 20 \sin^2 x + 4 = 0$

$\Rightarrow \sin^2 x = \frac{2}{5} \text{ and } \cos^2 x = \frac{3}{5}$

$\therefore \tan^2 x = \frac{2}{3} \text{ and } \frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

W.E-20: Let $f: (-1, 1) \rightarrow R$ be such that

$f(\cos 4\theta) = \frac{2}{2 - \sec^2 \theta}$ for $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$.

Then the value(s) of $f\left(\frac{1}{3}\right)$ is/are **[IIT 2012]**

A) $1 - \sqrt{\frac{3}{2}}$ B) $1 + \sqrt{\frac{3}{2}}$

C) $1 - \sqrt{\frac{2}{3}}$ D) $1 + \sqrt{\frac{2}{3}}$

Ans: A,B

Sol: For $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Let $\cos 4\theta = \frac{1}{3}$

$\Rightarrow \cos 2\theta = \pm \sqrt{\frac{1 + \cos 4\theta}{2}} = \pm \sqrt{\frac{2}{3}}$

$f\left(\frac{1}{3}\right) = \frac{2}{2 - \sec^2 \theta} = \frac{2 \cos^2 \theta}{2 \cos^2 \theta - 1} = 1 + \frac{1}{\cos 2\theta}$

$f\left(\frac{1}{3}\right) = 1 - \sqrt{\frac{3}{2}} \text{ or } 1 + \sqrt{\frac{3}{2}}.$

LEVEL-V

SINGLE ANSWER QUESTIONS

1. If $x = \sec \theta + \tan \theta, y = \sec \theta - \cot \theta$ then $y =$

A) $\frac{1+x}{1-x}$ B) $\frac{1-x}{1+x}$ C) $\frac{x+1}{x-1}$ D) $\frac{x-1}{x+1}$

2. $0 \leq a \leq 3, 0 \leq b \leq 3$ and the equation $x^2 + 4 + 3 \cos(ax+b) = 2x$ has at least one solution then the value of $(a+b)$

A) $\frac{\pi}{2}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{3}$ D) π

3. If $\frac{\sin^4 \theta}{a} + \frac{\cos^4 \theta}{b} = \frac{1}{a+b}$

then $\frac{\sin^8 \theta}{a^3} + \frac{\cos^8 \theta}{b^3} =$

A) $\frac{1}{(a+b)^3}$ B) $\frac{1}{(a+b)^2}$

C) $a+b$ D) $\frac{1}{a+b}$

4. If $\sin^3 x \sin 3x = \sum_{m=0}^6 c_m \cos^m x$, where $c_0, c_1, \dots, c_6$ are constants, then

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- A) $c_0 + c_2 + c_4 + c_6 = 0$
 B) $c_1 + c_3 + c_5 = 0$
 C) $2c_2 + 3c_6 = 0$
 D) $c_4 + 2c_6 = 0$
5. The minimum value of $2\cos\theta + \frac{1}{\sin\theta} + \sqrt{2}\tan\theta$ in $\left(0, \frac{\pi}{2}\right)$ is
 A) $2 + \sqrt{2}$ B) $3\sqrt{2}$ C) $2\sqrt{3}$ D) $3 + \sqrt{2}$
6. If $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$, $0 < \theta < \frac{3\pi}{4}$
 then $\sin\left(\theta + \frac{\pi}{4}\right) =$
 A) $\frac{1}{\sqrt{2}}$ B) $-\frac{1}{\sqrt{2}}$ C) $\frac{1}{2\sqrt{2}}$ D) $-\frac{1}{2\sqrt{2}}$
7. If $\tan \alpha = \frac{1}{\sqrt{x(x^2 + x + 1)}}$,
 $\tan \beta = \frac{\sqrt{x}}{\sqrt{x^2 + x + 1}}$ and
 $\tan \gamma = \sqrt{x^{-3} + x^{-2} + x^{-1}}$ then $\alpha + \beta =$
 A) γ B) 2γ C) $-\gamma$ D) $\gamma/2$
8. $(\sin^8 75^\circ - \cos^8 75^\circ) =$
 A) 1 B) $\frac{3\sqrt{3}}{8}$ C) $\frac{3\sqrt{3}}{16}$ D) $\frac{7\sqrt{3}}{16}$
9. $\sin 20^\circ (4 + \sec 20^\circ) =$
 A) $\frac{1}{2}$ B) $\sqrt{2}$ C) $\sqrt{3}$ D) 1
10. $3\tan^6 10^\circ - 27\tan^4 10^\circ + 33\tan^2 10^\circ =$
 A) 0 B) 1 C) 2 D) 3
11. If $4n\alpha = \pi$, then the numerical value of $\tan \alpha \tan 2\alpha \tan 3\alpha \dots \tan (2n-1)\alpha$ is equal to
 A) -1 B) 0

- C) 1 D) 2
12. If $\sec^4 \theta + \sec^2 \theta = 10 + \tan^4 \theta + \tan^2 \theta$
 then $\sin^2 \theta =$
 A) $\frac{2}{3}$ B) $\frac{3}{4}$ C) $\frac{4}{5}$ D) $\frac{5}{6}$
13. In $\triangle ABC$, $\angle A = \frac{\pi}{4}$ and $\tan B \tan C = P$
 then all possible values of P is
 A) $(-\infty, 3 - 2\sqrt{2}) \cup (3 + 2\sqrt{2}, \infty)$
 B) $(-\infty, 3 - 2\sqrt{2}] \cup [3 + 2\sqrt{2}, \infty)$
 C) $(3 - 2\sqrt{2}, 3 + 2\sqrt{2})$
 D) $[3 - 2\sqrt{2}, 3 + 2\sqrt{2}]$
14. $\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} =$
 A) $\sqrt{7}/2$ B) $7/2$
 C) $-\sqrt{7}/2$ D) $-7/2$
15. $\sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{4\pi}{7} =$
 A) $\frac{\sqrt{7}}{8}$ B) $\frac{\sqrt{7}}{4}$ C) $\frac{\sqrt{7}}{2}$ D) $\sqrt{7}$
16. If $\tan\left(x + \frac{\pi}{4}\right) = a$ then $\sec^2 x =$
 A) $1 + a^2$ B) $\frac{4a}{(a+1)^2}$
 C) $\frac{2(a^2+1)}{(a+1)^2}$ D) $\left(\frac{a-1}{a+1}\right)^2$
17. If α, β, γ do not differ by a multiple of π and if
 $\frac{\cos(\alpha + \theta)}{\sin(\beta + \gamma)} = \frac{\cos(\beta + \theta)}{\sin(\gamma + \alpha)} = \frac{\cos(\gamma + \theta)}{\sin(\alpha + \beta)} = k$.
 Then k equals
 A) ± 2 B) $\pm \frac{1}{2}$ C) 0 D) ± 1

18. If A, B, C are in A.P and $B = \frac{\pi}{4}$
then $\tan A \tan B \tan C =$
A) 1 B) 2 C) 3 D) 4
19. If $\theta = \frac{\pi}{2^n + 1}$ then
 $\cos \theta \cos 2\theta \cos 2^2 \theta \dots \cos 2^{n-1} \theta$ is equal to
A) $\frac{1}{2^n}$ B) $\cos \theta$ C) 2 D) 2^n
20. If ABC is a triangle and $\tan \frac{A}{2}, \tan \frac{B}{2}, \tan \frac{C}{2}$
are in H.P then the minimum value of $\cot \frac{B}{2}$
is equal to
A) $-\sqrt{3}$ B) $\sqrt{3}$ C) 2 D) -2
21. $\sum_{r=1}^{10} \cos^3 \frac{r\pi}{3} =$
A) $-\frac{1}{8}$ B) $-\frac{7}{8}$ C) $-\frac{9}{8}$ D) $\frac{1}{8}$
22. If $\sin \frac{\pi}{18} \sin \frac{5\pi}{18} \sin \frac{7\pi}{18} =$
A) 8 B) $\frac{1}{8}$ C) $1/7$ D) 6
23. The value of
 $\cos y \cos(\pi/2 - x) - \cos(\pi/2 - y) \cos x +$
 $\sin y \cos(\pi/2 - x) + \cos x \sin(\pi/2 - y)$
is zero if
A) $x = 0$ B) $y = 0$ C) $x = y$
D) $x = n\pi - \pi/4 + y (n \in I)$
24. If $\tan \theta_1, \tan \theta_2, \tan \theta_3$ and $\tan \theta_4$ are the
roots of the equation $x^4 - x^3 \sin 2\beta$
 $+ x^2 \cos 2\beta - x \cos \beta - \sin \beta = 0$
then $\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4)$ is equal to
A) $\sin \beta$ B) $\cos \beta$ C) $\tan \beta$ D) $\cot \beta$
25. If $\frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta - \cos \theta} - \frac{\cos \theta}{\sqrt{1 + \cot^2 \theta}}$
- $-2 \tan \theta \cot \theta = -1, \theta \in [0, 2\pi]$ then
A) $\theta \in (0, \pi/2) - \{\pi/4\}$
B) $\theta \in \left(\frac{\pi}{2}, \pi\right) - \{3\pi/4\}$
C) $\theta \in \left(\pi, \frac{3\pi}{2}\right) - \{5\pi/4\}$
D) $\theta \in (0, \pi) - \{\pi/4, \pi/2\}$
26. If $\frac{\tan(\alpha + \beta - \gamma)}{\tan(\alpha - \beta + \gamma)} = \frac{\tan \gamma}{\tan \beta} (\beta \neq \gamma)$ then
 $\sin 2\alpha + \sin 2\beta + \sin 2\gamma =$
A) 0 B) 1 C) 2 D) $1/2$
27. $\cos 56^\circ + \cos 58^\circ - \cos 66^\circ$
 $- 4 \cos 28^\circ \cos 29^\circ \sin 33^\circ =$
A) 0 B) 1 C) -1 D) 2
28. If $\frac{x}{y} = \frac{\cos A}{\cos B}$ then $\frac{x \tan A + y \tan B}{x + y} =$
A) $\tan \frac{A+B}{2}$ B) $\tan \frac{A-B}{2}$
C) $\cot \frac{A+B}{2}$ D) $\cot \frac{A-B}{2}$
29. If $\sin x + \operatorname{cosec} x + \tan y + \cot y = 4$ where x
and $y \in \left(0, \frac{\pi}{2}\right) \tan \frac{y}{2}$ is a root of the
equation
A) $\alpha^2 + 2\alpha + 1 = 0$ B) $\alpha^2 + 2\alpha - 1 = 0$
C) $2\alpha^2 - 2\alpha - 1 = 0$ D) $\alpha^2 - \alpha - 1 = 0$
30. An integral value of a for which there is a
solution of the equation
 $a \cos x + \cot x + 1 = \operatorname{cosec} x$ is ($\sin 2x \neq 0$)
A) 1 B) 2 C) 3 D) 4
31. If $y = (1 + \tan A)(1 - \tan B)$
where $A - B = \frac{\pi}{4}$ then $(y + 1)^{y+1} =$
A) 27 B) 9 C) 81 D) 4
32. Let A, B, C be three angles such that $A = \frac{\pi}{4}$
and $\tan B \tan C = p$. Then, all possible values
of p such that A, B, C are the angles of a

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triangle is

$$A) \leq (\sqrt{2} + 1)^2 \quad B) \geq (\sqrt{2} + 1)^2$$

$$C) > (\sqrt{2} + 1)^2 \quad D) < (\sqrt{2} + 1)^2$$

33. If $\tan \alpha = \frac{P}{q}$ where $\alpha = 2\beta$, α being an acute

angle then $\frac{1}{2}[p \operatorname{cosec} 2\beta + q \sec 2\beta]$ is equal to

$$A) \sqrt{p^2 + q^2} \quad B) \sqrt{p^2 - q^2}$$

$$C) \sqrt{2p^2 + q^2} \quad D) 2\sqrt{p^2 + q^2}$$

34. If $\frac{\cos x}{a} = \frac{\cos(x+\theta)}{b} = \frac{\cos(x+2\theta)}{c} = \frac{\cos(x+3\theta)}{d}$

then $\frac{a+c}{b+d}$ is equal to

$$A) a/d \quad B) c/d \quad C) b/c \quad D) d/a$$

35. If $\sqrt{2} \cos A = \cos B + \cos^3 B$ and $\sqrt{2} \sin A = \sin B - \sin^3 B$ then $\sin(A-B) =$

$$A) \pm 1 \quad B) \pm \frac{1}{2} \quad C) \pm \frac{1}{3} \quad D) \pm \frac{1}{4}$$

36. For any real θ , the maximum value of $\cos^2(\cos \theta) + \sin^2(\sin \theta)$ is

$$A) 1 \quad B) 1 + \sin^2 1 \\ C) 1 + \cos^2 1 \quad D) \text{does not exist}$$

37. The value of $\cot \frac{7\pi}{16} + 2 \cot \frac{3\pi}{8} + \cot \frac{15\pi}{16}$ is

$$A) 4 \quad B) 2 \quad C) -2 \quad D) -4$$

38. If $0 < \alpha < \frac{\pi}{6}$ then $\alpha(\operatorname{cosec} \alpha)$ is

$$A) \text{less than } \frac{\pi}{6} \quad B) \text{greater than } \frac{\pi}{6}$$

$$C) \text{less than } \frac{\pi}{3} \quad D) \text{greater than } \frac{\pi}{3}$$

39. Let $P = \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$ and

$Q = \{\theta : \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$ be two sets. Then [IIT-2011]

$$(A) P \subset Q \text{ and } P \neq \emptyset$$

$$(B) Q \not\subset P$$

$$(C) P \not\subset Q$$

$$(D) P = Q$$

40. The maximum value of $(\cos \alpha_1)(\cos \alpha_2) \dots (\cos \alpha_n)$ under the restrictions $0 \leq \alpha_1, \alpha_2, \dots, \alpha_n \leq \frac{\pi}{2}$ and

$$\cot \alpha_1 \cot \alpha_2 \dots \cot \alpha_n = 1 \text{ is [IIT 2001]}$$

$$A) \frac{1}{2^{n/2}} \quad B) \frac{1}{2^n} \quad C) \frac{1}{2n} \quad D) 1$$

41. If $\tan \theta - \cot \theta = a$ and $\cos \theta - \sin \theta = b$ then $(a^2 + 4)(b^2 - 1)^2$

$$A) 1 \quad B) 3 \quad C) 4 \quad D) 5$$

42. The maximum value of the expression

$$\left| \sqrt{(\sin^2 x + 2a^2)} - \sqrt{(2a^2 - 1 - \cos^2 x)} \right|$$

where a and x are real numbers is

$$A) \sqrt{3} \quad B) \sqrt{2} \quad C) 1 \quad D) \sqrt{5}$$

MULTIPLE ANSWER QUESTIONS

43. If the equation $\sin x(\sin x + \cos x) = k$ has real solutions then k may lie in the interval

$$A) \left[0, \frac{\sqrt{2}+1}{2}\right] \quad B) [2-\sqrt{3}, 2+\sqrt{3}]$$

$$C) [0, 2-\sqrt{3}] \quad D) \left[\frac{1-\sqrt{2}}{2}, \frac{1+\sqrt{2}}{2}\right]$$

44. If $\cos x + \cos y + \cos z = 0 = \sin x + \sin y + \sin z$

then the possible value of $\cos\left(\frac{x-y}{2}\right) =$

[IIT 2013]

$$A) \frac{1}{2} \quad B) -\frac{1}{2} \quad C) 1 \quad D) -1$$

45. If $x = \sin(\alpha - \beta)\sin(\gamma - \delta)$

$$y = \sin(\beta - \gamma)\sin(\alpha - \delta) \text{ and}$$

$$z = \sin(\gamma - \alpha)\sin(\beta - \delta) \text{ then}$$

$$A) x + y + z = 0 \quad B) x + y - z = 0$$

$$C) y + z - x = 0 \quad D) x^3 + y^3 + z^3 = 3xyz$$

46. If $\frac{\tan 3A}{\tan A} = k$ ($k \neq 1$) then

$$A) \frac{\cos A}{\cos 3A} = \frac{k^2 - 1}{2k} \quad B) \frac{\sin 3A}{\sin A} = \frac{2k}{k-1}$$

$$C) k < \frac{1}{3} \quad D) k > 3$$

47. If A and B are acute angles such that $\sin A = \sin^2 B$, $2 \cos^2 A = 3 \cos^2 B$ then

$$A) A = \pi/6 \quad B) A = \pi/2 \\ C) B = \pi/4 \quad D) B = \pi/3$$

48. For $0 < \phi < \pi/2$, if

$$x = \sum_{n=0}^{\infty} \cos^{2n} \phi, y = \sum_{n=0}^{\infty} \sin^{2n} \phi \text{ and}$$

$$z = \sum_{n=0}^{\infty} \cos^{2n} \phi \sin^{2n} \phi \text{ then } xyz =$$

$$(A) xy + z \quad (B) xz + y \\ (C) x + y + z \quad (D) yz + x$$

49. Let

$$f(x) = a_1 \cos(\alpha_1 + x) + a_2 \cos(\alpha_2 + x) + \dots + a_n \cos(\alpha_n + x).$$

If $f(x)$ vanishes for $x = 0$ and $x = x_1$

(where $x_1 \neq k\pi$, $k \in \mathbb{Z}$) then

$$A) a_1 \cos \alpha_1 + a_2 \cos \alpha_2 + \dots + a_n \cos \alpha_n = 0$$

$$B) a_1 \sin \alpha_1 + a_2 \sin \alpha_2 + \dots + a_n \sin \alpha_n = 0$$

$$C) f(x) = 0 \text{ has only two solutions } 0, x_1$$

$$D) f(x) \text{ is identically zero } \forall x$$

50. For $\alpha = \pi/7$ which of the following hold(s) good?

$$A) \tan \alpha \tan 2\alpha \tan 3\alpha = \tan 3\alpha - \tan 2\alpha - \tan \alpha$$

$$B) \operatorname{cosec} \alpha = \operatorname{cosec} 2\alpha + \operatorname{cosec} 4\alpha$$

$$C) \cos \alpha - \cos 2\alpha + \cos 3\alpha = 1/2$$

$$D) 8 \cos \alpha \cos 2\alpha \cos 4\alpha = 1$$

51. Which of the following is/are correct?

$$A) (\tan x)^{\ln(\sin x)} > (\cot x)^{\ln(\sin x)} \forall x \in \left(0, \frac{\pi}{4}\right)$$

$$B) 4^{\ln \operatorname{cosec} x} < 5^{\ln \operatorname{cosec} x} \forall x \in \left(0, \frac{\pi}{2}\right)$$

$$C) \left(\frac{1}{2}\right)^{\ln(\cos x)} < \left(\frac{1}{3}\right)^{\ln(\cos x)} \forall x \in \left(0, \frac{\pi}{2}\right)$$

$$D) 2^{\ln(\tan x)} > 2^{\ln(\sin x)} \forall x \in \left(0, \frac{\pi}{2}\right)$$

COMPREHENSION TYPE

(P) Given, $\cos 2^m \theta \cos 2^{m+1} \theta \dots \cos 2^n \theta$

$$= \frac{\sin 2^{n+1} \theta}{2^{n-m+1} \sin 2^m \theta}$$

where $2^m \theta \neq k\pi$, $n, m, k \in \mathbb{I}$ solve the following

$$52. \sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14} \text{ is equal to}$$

$$A) \frac{1}{64} \quad B) -\frac{1}{64} \quad C) \frac{1}{8} \quad D) -\frac{1}{8}$$

$$53. \cos 2^3 \frac{\pi}{10} \cos 2^4 \frac{\pi}{10} \cos 2^5 \frac{\pi}{10} \dots \cos 2^{10} \frac{\pi}{10} \text{ is equal to}$$

$$A) \frac{1}{128} \quad B) \frac{1}{256}$$

$$C) \frac{1}{512} \sin \frac{\pi}{10} \quad D) \frac{\sqrt{5}-1}{512} \sin \frac{3\pi}{10}$$

$$54. \cos \frac{\pi}{11} \cos \frac{2\pi}{11} \cos \frac{3\pi}{11} \dots \cos \frac{10\pi}{11} \text{ is equal to}$$

$$A) -\frac{1}{32} \quad B) \frac{1}{512} \quad C) \frac{1}{1024} \quad D) -\frac{1}{1024}$$

$$(P) \text{ In a } \triangle ABC, \text{ if } \cos A \cos B \cos C = \frac{\sqrt{3}-1}{8}$$

$$\text{and } \sin A \sin B \sin C = \frac{3+\sqrt{3}}{8} \text{ then}$$

On the basis of above information, answer the following questions:

55. The value of $\tan A + \tan B + \tan C$ is

$$(A) \frac{3+\sqrt{3}}{\sqrt{3}-1} \quad (B) \frac{\sqrt{3}+4}{\sqrt{3}-1} \\ (C) \frac{6-\sqrt{3}}{\sqrt{3}-1} \quad (D) \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-1}$$

56. The value of $\tan A \tan B + \tan B \tan C + \tan C \tan A$ is

$$(A) 5-4\sqrt{3} \quad (B) 5+4\sqrt{3}$$

(C) $6 + \sqrt{3}$ (D) $6 - \sqrt{3}$

57. The angles of $\triangle ABC$ are

- (A) $45^\circ, 30^\circ, 105^\circ$ (B) $45^\circ, 60^\circ, 75^\circ$
(C) $45^\circ, 45^\circ, 90^\circ$ (D) $45^\circ, 60^\circ, 70^\circ$.

(P) If $7\theta = (2n+1)\pi$, when $n = 0, 1, 2, 3, 4, 5, 6$ then on the basis of above information, answer the following questions

58. The equation whose roots are $\cos \pi/7$, $\cos 3\pi/7$, $\cos 5\pi/7$ is

- (A) $8x^3 + 4x^2 + 4x + 1 = 0$
(B) $8x^3 - 4x^2 - 4x + 1 = 0$
(C) $8x^3 - 4x^2 - 4x - 1 = 0$
(D) $8x^3 + 4x^2 + 4x - 1 = 0$

59. The value of $\sec \pi/7 + \sec 3\pi/7 + \sec 5\pi/7$ is

- (A) 4 (B) -4 (C) 3 (D) -3.

(P) Consider the function defined by

$$f(x) = \frac{\sin x}{\sqrt{1 + \tan^2 x}} + \frac{\cos x}{\sqrt{1 + \cot^2 x}}$$

for permissible values. Then

60. Which of the following is true?

- (A) $f(x) = \sin 2x$ for all x
(B) $f(x) = \sin x |\cos x| + \cos x |\sin x|$ for all x
(C) $f(x) = 0$ for $\frac{\pi}{2} < x < \pi$
(D) $f(x) = \sin x$ for all x

61. $f(x) = 0$ (n is an integer) for

- (A) $2n\pi < x < 2n\pi + \frac{\pi}{2}$
(B) $(2n+1)\frac{\pi}{2} < x < (2n+2)\frac{\pi}{2}$
(C) $(2n+1)\pi < x < (2n+2)\pi$
(D) $0 < x < 2n\pi$

62. $f(x) < 0$ for all x in (n is any integer)

- (A) $2n\pi < x < 2n\pi + \frac{\pi}{2}$
(B) $2n\pi < x < (2n+1)\pi$
(C) $(2n+1)\pi < x < (4n+3)\frac{\pi}{2}$
(D) $0 < x < 2n\pi$

MATRIX MATCHING QUESTIONS

63. Match the following

COLUMN-I

A) When θ is fixed constant then the maximum value of $\{\cos(2A + \theta) + \cos(2B + \theta)\}$

B) The maximum value of $\{\cos 2A + \cos 2B\}$ where $(A+B)$ is constant and $A, B \in (0, \pi/2)$, is

C) The minimum value of $\{\sec 2A + \sec 2B\}$, where $(A+B)$ is constant and $A, B \in (0, \pi/4)$, is

D) The minimum value of

$\sqrt{\{\tan \theta + \cot \theta - 2 \cos 2(A+B)\}}$, where A, B are constant and $\theta \in (0, \pi/2)$ is

COLUMN-II

p) $2 \sin(A+B)$

q) $2 \sec(A+B)$

r) $2 \cos(A+B)$

s) $2 \cos(A-B)$

64. Match the following:

COLUMN-I

A) If maximum and minimum values of

$\frac{7 + 6 \tan \theta - \tan^2 \theta}{(1 + \tan^2 \theta)}$ for all real values of

$\theta \neq (2n+1)\frac{\pi}{2}$ are λ and μ respectively then

B) If maximum and minimum values of

$5 \cos \theta + 3 \cos\left(\theta + \frac{\pi}{3}\right) + 3$ for all real values of

θ are λ and μ respectively then

C) If maximum and minimum values of

$1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2 \cos\left(\frac{\pi}{4} - \theta\right)$ for all real

values of θ are λ and μ respectively then

COLUMN-II

p) $\lambda + \mu = 2$

q) $\lambda - \mu = 6$

r) $\lambda + \mu = 6$

s) $\lambda - \mu = 10$

t) $\lambda - \mu = 14$

65. Match the following:

COLUMN-I

A) In an acute angled triangle ABC, the least

values of $\sum \sec A$ and $\sum \tan^2 A$ are λ and μ respectively then

B) In a triangle ABC, the least values of

$\sum \operatorname{cosec}\left(\frac{A}{2}\right)$ and $\sum \sec^2\left(\frac{A}{2}\right)$ are

λ and μ respectively then

C) In a triangle ABC, the least values of

$\Pi \operatorname{cosec}\left(\frac{A}{2}\right)$ and $\sum \operatorname{cosec}^2 A$ are λ and μ

respectively then

COLUMN-II

p) $\lambda - \mu = 2$

q) $\mu - \lambda = 3$

r) $\lambda - \mu = 4$

s) $3\lambda - 2\mu = 0$

t) $2\lambda - 3\mu = 0$

ASSERTION-REASONING QUESTIONS

This section contains 2 questions. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which ONLY ONE is correct.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False

(D) Statement -1 is False, Statement-2 is True.

66. STATEMENT - 1:

The minimum value of the expression $\sin \alpha + \sin \beta + \sin \gamma$ where α, β, γ are real numbers such that $\alpha + \beta + \gamma = \pi$, is negative because

STATEMENT - 2: α, β, γ are angles of a triangle

67. STATEMENT-1: The equation $\sin^2 x + \cos^2 y = 2 \sec^2 z$ is only solvable if $\sin x = 1, \cos y = 1$ and $\sec z = 1$ where $x, y, z \in \mathbb{R}$ because

STATEMENT - 2: Maximum value of $\sin x$ and $\cos y$ is 1 and minimum value of $\sec z$ is 1.

68. STATEMENT-1 : The maximum and minimum values of the function

$f(x) = \frac{1}{3 \sin x + 4 \cos x - 2}$ does not exist

STATEMENT - 2: The given function is an unbounded function.

INTEGER QUESTIONS

69. If $\tan x + \tan 2x + \tan 3x = \tan x \tan 2x \tan 3x$ then $|\sin 3x + \cos 3x| =$

70. $16 \left(\cos \theta - \cos \frac{\pi}{8} \right) \left(\cos \theta - \cos \frac{3\pi}{8} \right) \left(\cos \theta - \cos \frac{5\pi}{8} \right) \left(\cos \theta - \cos \frac{7\pi}{8} \right) = \lambda \cos 4\theta$ then the value of λ is

71. Given that $f(n\theta) = \frac{2 \sin 2\theta}{\cos 2\theta - \cos 4n\theta}$ and $f(\theta) + f(2\theta) + f(3\theta) + \dots + f(n\theta) = \frac{\sin \lambda \theta}{\sin \theta \sin \mu \theta}$ then the value of $\mu - \lambda$, is

72. If $4 \sin 27^\circ = \sqrt{\alpha} + \sqrt{\beta}$ then the sum of the digits in $(\alpha + \beta - \alpha\beta + 2)^4$

73. If $\sin^2 A = x$ and then the

$\prod_{r=1}^4 \sin(rA) = ax^2 + bx^3 + cx^4 + dx^5$ value of $a + b + c + d$ must be

74. If $\cot(\theta - \alpha), 3 \cot \theta, \cot(\theta + \alpha)$ are in A.P and θ is not an integral multiple of $\frac{\pi}{2}$ then

the value of $\frac{4 \sin^2 \theta}{3 \sin^2 \alpha}$ must be

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75. If $A + B + C = \pi$ and

$$\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = \lambda \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right),$$

then the value of λ must be

76. If $a \tan \alpha + \sqrt{a^2 - 1} \tan \beta + \sqrt{a^2 + 1} \tan \gamma = 2a$, where a is constant and α, β, γ are variable angles, by using $\vec{U} \cdot \vec{V} \leq |\vec{U}| |\vec{V}|$

The least value of $(\tan^2 \alpha + \tan^2 \beta + \tan^2 \gamma)$ is

77. The maximum value of the expression

$$\frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta} \text{ is [IIT 2010]}$$

78. The positive integer value of $n > 3$ satisfying the equation

$$\frac{1}{\sin\left(\frac{\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)} + \frac{1}{\sin\left(\frac{3\pi}{n}\right)} \text{ is}$$

[IIT 2011]

SUBJECTIVE QUESTIONS

79. Show that $\cos(\sin \theta) > \sin(\cos \theta)$, $0 \leq \theta \leq \frac{\pi}{2}$

80. If $e^{\frac{\pi}{2}} < \theta < e^{\frac{\pi}{2}}$ then P.T $\cos(\ln \theta) > \ln(\cos \theta)$?

81. Prove that $\sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2n-1)\theta = \frac{\sin^2 n\theta}{\sin \theta}$

82. In $\triangle ABC$, P.T $\tan A + \tan B + \tan C \geq 3\sqrt{3}$, where A, B, C are acute angles.

83. In $\triangle ABC$ prove that

$$\operatorname{cosec} \frac{A}{2} + \operatorname{cosec} \frac{B}{2} + \operatorname{cosec} \frac{C}{2} \geq 6$$

84. prove that $\frac{3 - \tan^2 \frac{\pi}{7}}{1 - \tan^2 \frac{\pi}{7}} = 4 \cos \frac{\pi}{7}$

85. Show that the number of solutions of the equation $[\sin x] = [1 + \sin x] + [1 - \cos x]$ is Zero (where $[.]$ is step x , $0 \leq x \leq 2\pi$)

86. P.T $\frac{1}{\sin 45^\circ \sin 46^\circ} + \frac{1}{\sin 47^\circ \sin 48^\circ} + \dots + \frac{1}{\sin 133^\circ \sin 134^\circ} = \operatorname{cosec} 1^\circ$

87. In $\triangle ABC$, if $\cot \theta = \cot A + \cot B + \cot C$, then prove that

$$\sin(A-\theta) \sin(B-\theta) \sin(C-\theta) = \sin^3 \theta$$

88. In $\triangle ABC$, $\cot A + \cot B + \cot C = 0$ then P.T $\cos A \cos B \cos C = -1$

89. Show that sum of the series $\operatorname{cosec} \theta + \operatorname{cosec} 2\theta + \operatorname{cosec} 4\theta + \dots$ n terms $= \cot \frac{\theta}{2} - \cot 2^{n-1} \theta$

90. P.T $\tan^2 \frac{\pi}{7} + \tan^2 \frac{2\pi}{7} + \tan^2 \frac{3\pi}{7} = 21$

91. In triangle ABC given $\tan(A-B) + \tan(B-C) + \tan(C-A) = 0$ then show that the triangle is isosceles triangle

92. If $\frac{\cos \alpha}{\cos \beta} + \frac{\sin \alpha}{\sin \beta} = -1$ then show that the value of $\frac{\cos^3 \beta}{\cos \alpha} + \frac{\sin^3 \beta}{\sin \alpha} = 1$

93. Prove that $\sum_{r=1}^n \tan r\alpha \tan(r+1)\alpha = \cot \alpha \tan(n+1)\alpha - n - 1$

94. In $\triangle ABC$, $3 \sin A + 4 \cos B = 6$, $4 \sin B + 3 \cos A = 1$ then show that the number of possible values of $\angle C$ is one

95. Let $a^2 + b^2 = \alpha^2 + \beta^2 = 2$ then show that the maximum value of $S = (1-a)(1-b) + (1-\alpha)(1-\beta)$ is 8

LEVEL-V KEY

SINGLE ANSWER QUESTIONS

- | | | | |
|-------|-------|-------|-------|
| 1) D | 2) D | 3) A | 4) A |
| 5) B | 6) C | 7) A | 8) D |
| 9) C | 10) B | 11) C | 12) C |
| 13) B | 14) A | 15) A | 16) C |

- 17) D 18) A 19) A 20) B
 21) C 22) B 23) D 24) D
 25) D 26) A 27) C 28) A
 29) B 30) C 31) A 32) B
 33) A 34) C 35) C 36) B
 37) D 38) C 39) D 40) A
 41) C 42) B

MULTIPLE ANSWER QUESTIONS

- 43) A,C,D 44) A,B
 45) A,D 46) B,C,D
 47) A,C 48) A, C
 49) A, B, D 50) (A, B,C)
 51) A,B,C,D

COMPREHENSION QUESTIONS

- 52) C 53) B 54) D 55) A
 56) B 57) B 58) B 59) A
 60) C 61) B 62) C

MATRIX MATCH QUESTIONS

- 63) $A \rightarrow s; B \rightarrow r; C \rightarrow q; D \rightarrow p$
 64) $A \rightarrow r, s; B \rightarrow r, t; C \rightarrow p, q$
 65) $A \rightarrow q, s; B \rightarrow p, t; C \rightarrow r$

ASSERTION-REASONING QUESTIONS

- 66) C 67) C 68) A

INTEGER QUESTIONS

- 69) 1 70) 2 71) 1 72) 4
 73) 0 74) 2 75) 8 76) 4
 77) 2 78) 7

LEVEL - V - HINTS

SINGLE ANSWER QUESTIONS

01. $y = \frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$

$$x = \frac{1 + \sin \theta}{\cos \theta}$$

$$\Rightarrow \frac{x+1}{x-1} = \frac{1 + \cos \theta + \sin \theta}{1 - \cos \theta + \sin \theta} = \cot \frac{\theta}{2}$$

02. $x^2 - 2x + 4 = -3 \cos(ax + b)$

$$\Rightarrow (x-1)^2 + 3 = -3 \cos(ax + b)$$

As $-1 \leq \cos(ax + b) \leq 1$ and $(x-1)^2 \geq 0$

$\therefore$ equation (i) is only possible if,
 $\cos(ax + b) = -1$ and $(x-1) = 0$

so $a + b = \pi, 3\pi, 5\pi, \dots$

and $3\pi > 6$

$$\Rightarrow a + b = \pi \text{ where } a + b \leq 6$$

03. $\frac{\sin^4 \theta}{a} + \frac{(1 - \sin^2 \theta)^2}{b} = \frac{1}{a+b}$

$$(a+b)^2 \sin^4 \theta - 2a(a+b) \sin^2 \theta + a^2 = 0$$

$$(a+b) \sin^2 \theta = a$$

$$\therefore \sin^2 \theta = \frac{a}{a+b}, \cos^2 \theta = \frac{b}{a+b}$$

Now

$$\frac{\sin^8 \theta}{a^3} + \frac{\cos^8 \theta}{b^3} = \frac{1}{a^3} \left(\frac{a}{a+b} \right)^4 + \frac{1}{b^3} \left(\frac{b}{a+b} \right)^4$$

$$= \frac{a+b}{(a+b)^4} = \frac{1}{(a+b)^3}$$

04. $\sin^3 x \sin 3x = \sin^3 x (3 \sin x - 4 \sin^3 x)$
 $= \sin^4 x (3 - 4 \sin^2 x)$

$$= (1 - \cos^2 x)^2 (3 - 4(1 - \cos^2 x))$$

$$= (1 - 2 \cos^2 x + \cos^4 x)(4 \cos^2 x - 1)$$

$$= 4 \cos^6 x - 9 \cos^4 x + 6 \cos^2 x - 1$$

So that

$$c_0 = -1, c_2 = 6, c_4 = -9, c_6 = 4$$

and $c_1 = c_3 = c_5 = 0$

$$\Rightarrow c_0 + c_2 + c_4 + c_6 = 0$$

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05. $AM \geq GM$

06. $\tan(\pi \cos \theta) = \tan\left(\frac{\pi}{2} - \pi \sin \theta\right)$
 $\cos \theta + \sin \theta = \frac{1}{2} \Rightarrow \sin\left(\theta + \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$

07. Calculate $\tan(\alpha + \beta)$

08. Given expression =
 $(\sin^2 75^\circ - \cos^2 75^\circ)(\sin^4 75^\circ + \cos^4 75^\circ)$
 and simplify.

09. $\frac{\sin 20^\circ}{\cos 20^\circ}(4 \cos 20^\circ + 1) = \frac{2 \sin 40^\circ + \sin 20^\circ}{\cos 20^\circ}$
 $= \frac{\sin 40^\circ + \sin 40^\circ + \sin 20^\circ}{\cos 20^\circ}$
 $= \frac{\sin 40^\circ + \cos 10^\circ}{\cos 20^\circ} = \frac{\sin 40^\circ + \sin 80^\circ}{\cos 20^\circ}$
 $= 2 \sin 60^\circ = \sqrt{3}$

10. Let $t = \tan 10^\circ, \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{3t - t^3}{1 - 3t^2} = \frac{1}{\sqrt{3}}$
 squaring, $3t^6 - 27t^4 + 33t^2 = 1$

11. $\therefore 4n\alpha = \pi$

$\therefore \tan(2n-1)\alpha = \tan(2n\alpha - \alpha) = \tan\left(\frac{\pi}{2} - \alpha\right) = \cot \alpha$

$\tan(2n-2)\alpha = \tan(2n\alpha - 2\alpha)$
 $= \tan\left(\frac{\pi}{2} - 2\alpha\right) = \cot 2\alpha$

.....

$\therefore \tan \alpha \tan 2\alpha \tan 3\alpha \dots \tan(2n-1)\alpha$

$(\tan \alpha \tan(2n-1)\alpha)(\tan 2\alpha \tan(2n-2)\alpha) \dots \tan n\alpha$
 $= (1 \cdot 1 \dots 1) \tan n\alpha$
 $= \tan n\alpha = \tan \frac{\pi}{4} = 1$

12. $10 = \sec^4 \theta + \sec^2 \theta - \tan^4 \theta - \tan^2 \theta$
 $= \sec^2 \theta + \tan^2 \theta + 1 = 2 \sec^2 \theta$

$\cos^2 \theta = \frac{1}{5} \Rightarrow \sin^2 \theta = \frac{4}{5}$

13. $\tan B \tan C = P$

$\tan B \tan\left(\frac{3\pi}{4} - B\right) = P$

$\Rightarrow \tan^2 B + (1-P) \tan B + P = 0$

$\therefore \tan B$ is real $\Delta \geq 0$

14. Conceptual

15. $\sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{4\pi}{7}$
 $= \frac{1}{2} \left(\cos \frac{\pi}{7} - \cos \frac{3\pi}{7} \right) \sin \frac{4\pi}{7}$
 $= \frac{1}{4} \left[\sin \frac{5\pi}{7} + \sin \frac{3\pi}{7} - \left(\sin \pi + \sin \frac{\pi}{7} \right) \right]$
 $= \frac{1}{4} \left[\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} \right] = \frac{\sqrt{7}}{8}$

16. $\frac{1 + \tan x}{1 - \tan x} = a \Rightarrow \tan x = \frac{a-1}{a+1}$

$\sec^2 x = 1 + \tan^2 x = 1 + \left(\frac{a-1}{a+1} \right)^2 = \frac{2(a^2+1)}{(a+1)^2}$

17. Taking first two members, then

$2 \cos(\alpha + \theta) \sin(\gamma + \alpha) = 2 \cos(\beta + \theta) \sin(\beta + \gamma)$
 $\Rightarrow \sin(2\alpha + \theta + \gamma) - \sin(\theta - \gamma)$

$= \sin(2\beta + \theta + \gamma) - \sin(\theta - \gamma)$

$\Rightarrow \sin(2\alpha + \theta + \gamma) = \sin(2\beta + \theta + \gamma)$

$\Rightarrow 2\alpha + \theta + \gamma = n\pi + (-1)^n (2\beta + \theta + \gamma)$

for $n = 0$, $2\alpha + \theta + \gamma = 2\beta + \theta + \gamma$

$\therefore \alpha = \beta$

Similarly taking last two members, we get

$\beta = \gamma$

Hence, $\alpha = \beta = \gamma$

Also, take $n = 1, -1$

Then, we get $\alpha + \beta + \gamma + \theta = \frac{\pi}{2}, -\frac{\pi}{2}$

$$\therefore k = \frac{\cos(\alpha + \theta)}{\sin(\beta + \gamma)} = \frac{\cos\left(\frac{\pi}{2} - (\beta + \gamma)\right)}{\sin(\beta + \gamma)} = 1$$

and

$$k = \frac{\cos\left(-\frac{\pi}{2} - (\beta + \gamma)\right)}{\sin(\beta + \gamma)} = -1$$

Hence, $k = \pm 1$

$$18. \quad \tan\left(\frac{\pi}{4} - \theta\right) \tan \frac{\pi}{4} \tan\left(\frac{\pi}{4} + \theta\right) = 1$$

19. Conceptual

20. In a triangle ABC, $A+B+C=\pi$

$$\Rightarrow \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2} = \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} \dots (i)$$

But $\Rightarrow \tan \frac{A}{2}, \tan \frac{B}{2}, \tan \frac{C}{2}$ are in HP

$\Rightarrow \cot \frac{A}{2}, \cot \frac{B}{2}, \cot \frac{C}{2}$ are in AP

$$\cot \frac{A}{2} + \cot \frac{C}{2} = 2 \cot \frac{B}{2}$$

$$\cot \frac{A}{2} \cot \frac{C}{2} = 3$$

$$\Rightarrow GM \text{ of } \cot \frac{A}{2} \text{ and } \cot \frac{C}{2} = \sqrt{\cot \frac{A}{2} \cot \frac{C}{2}} = \sqrt{3}$$

$$\text{and AM of } \cot \frac{A}{2} \text{ and } \cot \frac{C}{2} = \frac{\cot \frac{A}{2} + \cot \frac{C}{2}}{2}$$

$$= \cot \frac{B}{2}$$

But $AM \geq GM$

$$\Rightarrow \cot \frac{B}{2} \geq \sqrt{3}$$

21. Conceptual

$$\begin{aligned} 22. \quad & \sin \frac{\pi}{18} \sin \frac{5\pi}{18} \sin \frac{7\pi}{18} \\ &= \sin 10^\circ \sin 50^\circ \sin 70^\circ \\ &= \frac{1}{2} [2 \sin 70^\circ \sin 10^\circ] \sin 50^\circ \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} [\cos 60^\circ - \cos 80^\circ] \sin 50^\circ \\ &= \frac{1}{4} \sin 50^\circ - \frac{1}{4} (2 \cos 80^\circ \sin 50^\circ) \\ &= \frac{1}{4} \sin 50^\circ - \frac{1}{4} (\sin 130^\circ - \sin 30^\circ) \\ &= \frac{1}{4} \sin 50^\circ - \frac{1}{4} \sin 50^\circ + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8} \end{aligned}$$

23. The given expression is equal to
 $\cos y \sin x - \sin y \cos x + \sin y \sin x$
 $+ \cos x \cos y = \sin(x-y) + \cos(x-y)$
 $\sin(x-y) + \cos(x-y)$

$$= \sqrt{2} \left[\frac{1}{\sqrt{2}} \sin(x-y) + \frac{1}{\sqrt{2}} \cos(x-y) \right]$$

$$= \sqrt{2} \sin \left[\frac{\pi}{4} + (x-y) \right] = 0 \text{ if}$$

$$\left[\frac{\pi}{4} + (x-y) \right] = n\pi \Rightarrow x-y = n\pi - \frac{\pi}{4}$$

24. from the given equation we get

$$S_1 = \tan \theta_1 + \tan \theta_2 + \tan \theta_3 + \tan \theta_4 = \sin 2\beta$$

$$S_2 = \sum \tan \theta_1 \tan \theta_2 = \cos 2\beta$$

$$S_3 = \sum \tan \theta_1 \tan \theta_2 \tan \theta_3 = \cos \beta$$

$$\text{and } S_4 = \tan \theta_1 \tan \theta_2 \tan \theta_3 \tan \theta_4 = -\sin \beta$$

$$\text{Now, } \tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) = \frac{S_1 - S_3}{1 - S_2 + S_4}$$

$$= \frac{\sin 2\beta - \cos \beta}{1 - \cos 2\beta - \sin \beta} = \frac{\cos \beta (2 \sin \beta - 1)}{\sin \beta (2 \sin \beta - 1)} = \cot \beta$$

25.

$$\frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta - \cos \theta} = \sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta,$$

$$\theta \neq \frac{\pi}{4}, \frac{5\pi}{4}$$

and

$$\frac{\cos \theta}{\sqrt{1 + \cot^2 \theta}} = \frac{\cos \theta}{|\sec \theta|} = \sin \theta \cos \theta, \forall \theta \in (0, \pi)$$

$$\text{and } -2 \tan \theta \cot \theta = -2, \theta \neq \frac{\pi}{2}$$

Hence, LHS=RHS

$$\text{but } \theta \neq \frac{\pi}{4}, \frac{5\pi}{4}, \frac{\pi}{2}$$

$$\text{Hence, } \theta \in (0, \pi) \sim \left\{ \frac{\pi}{4}, \frac{\pi}{2} \right\}.$$

26. Apply componendo and dividendo

$$\text{We get } \frac{\sin 2\alpha}{\sin 2(\beta - \gamma)} = \frac{\sin(\gamma + \beta)}{\sin(\gamma - \beta)}$$

$$\Rightarrow \sin 2(\beta - \gamma) \sin(\beta + \gamma)$$

$$+ \sin 2\alpha \sin(\beta - \gamma) = 0$$

$$\Rightarrow \sin(\beta - \gamma)(2 \cos(\beta - \gamma) \sin(\beta + \gamma) + \sin 2\alpha) = 0$$

$$\Rightarrow \sin(\beta - \gamma)(\sin 2\alpha + \sin 2\beta + \sin 2\gamma) = 0$$

27. $4 \cos 28^\circ \cos 29^\circ \sin 33^\circ$

$$= 2(\cos 1^\circ + \cos 57^\circ) \sin 33^\circ$$

$$= \sin 34^\circ + \sin 32^\circ + 1 - \sin 24^\circ$$

28. $\frac{\frac{x}{y} \tan A + \tan B}{\frac{x}{y} + 1} = \frac{\sin A + \sin B}{\cos A + \cos B}$

$$= \tan \frac{(A+B)}{2}$$

29. It is true when $x = \frac{\pi}{2}$, $y = \frac{\pi}{4}$

$$\tan y = 1 \Rightarrow \frac{2 \tan \frac{y}{2}}{1 - \tan^2 \frac{y}{2}} = 1$$

$$\Rightarrow \tan^2 \frac{y}{2} + 2 \tan \frac{y}{2} = 1$$

30. $\sin x + \cos x = 1 - \alpha \sin x \cos x$

$$\Rightarrow a^2 \sin^2 x \cos^2 x - 2(a+1) \sin x \cos x = 0$$

$$\Rightarrow \sin 2x \left(\frac{a^2}{2} \sin 2x - 2(a+1) \right) = 0$$

$$\Rightarrow \text{either } \sin 2x = 0 \forall a \in R \text{ or } \sin 2x = \frac{4(a+1)}{a^2}$$

for no solution $\left| \frac{4(a+1)}{a^2} \right| > 1$

$$\Rightarrow a \in (2 - \sqrt{2}, 2 + \sqrt{2}) \Rightarrow a \text{ cannot be equal to } 1, 2, 3$$

31. If $A - B = \frac{\pi}{4}$ then $(1 + \tan A)(1 - \tan B) = 2$

32. $\Rightarrow B + C = \frac{3\pi}{4} \Rightarrow 0 < B, C < \frac{3\pi}{4}$

$$\text{Also, } \tan B \tan C = p$$

$$\text{Let } y = \tan B, \tan C = p/y$$

$$\tan(B+C) = -1 \text{ leads quadratic in } y \text{ apply } \Delta \geq 0$$

$$p < 1 \text{ or } p \geq (\sqrt{2} + 1)^2$$

$$\text{combining Eq. (ii) and (iii) we get } p \geq (\sqrt{2} + 1)^2$$

33. $\frac{1}{2}[p \operatorname{cosec} 2\beta - q \sec 2\beta]$

$$\sqrt{\frac{p^2 + q^2}{2}} \left[\frac{p}{\sqrt{p^2 + q^2}} \operatorname{cosec} 2\beta - \frac{q}{\sqrt{p^2 + q^2}} \sec 2\beta \right]$$

and simplify

34. $\frac{a+c}{b+d} = \frac{\cos x + \cos(x+2\theta)}{\cos(x+\theta) + \cos(x+3\theta)}$

$$= \frac{2 \cos(x+\theta) \cos \theta}{2 \cos(x+2\theta) \cos \theta} = \frac{b}{c}$$

35. $\sqrt{2} \sin A \cos B - \sqrt{2} \cos A \sin B$

$$= (\sin B - \sin^3 B) \cos B - (\cos B + \cos^3 B) \sin B$$

$$= -\sin B \cos B \Rightarrow \sin(A-B) = -\frac{1}{2\sqrt{2}} \sin 2B$$

$$\text{and } 2 = \cos^2 B \sin^2 B + \cos^6 B$$

$$+ \sin^6 B + 2(\cos^4 B - \sin^4 B)$$

$$\Rightarrow \cos 2B = \frac{1}{3} \Rightarrow \sin^2 2B = \frac{8}{9}$$

36. Let $f(\theta) = \cos^2(\cos \theta) + \sin^2(\sin \theta)$

$$\because -1 \leq \cos \theta \leq 1 \text{ and } -1 \leq \sin \theta \leq 1$$

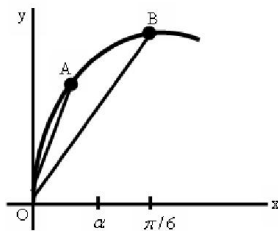
$$\therefore \cos 1 \leq \cos(\cos \theta) \leq 1 \text{ and } -\sin 1 \leq \sin(\sin \theta) \leq \sin 1$$

$$\therefore \cos^2 1 \leq \cos^2(\cos \theta) \leq 1 \text{ and } 0 \leq \sin^2(\sin \theta) \leq \sin^2 1$$

$$\therefore \text{Maximum value of } f(\theta) = 1 + \sin^2 1$$

$$\begin{aligned}
 37. \quad & \tan\left(\frac{\pi}{2} - \frac{7\pi}{16}\right) + 2 \tan\left(\frac{\pi}{2} - \frac{3\pi}{8}\right) - \cot\left(\pi - \frac{15\pi}{16}\right) \\
 &= \tan \frac{\pi}{16} - \cot \frac{\pi}{16} + 2 \tan \frac{\pi}{8} \\
 &= \frac{\sin \frac{\pi}{16}}{\cos \frac{\pi}{16}} - \frac{\cos \frac{\pi}{16}}{\sin \frac{\pi}{16}} + 2 \tan \frac{\pi}{8} \\
 &= \frac{2 \cos \frac{\pi}{8}}{\sin \frac{\pi}{8}} + 2 \frac{\sin \frac{\pi}{8}}{\cos \frac{\pi}{8}} = -\frac{4 \cos \frac{\pi}{4}}{\sin \frac{\pi}{4}} = -4
 \end{aligned}$$

38. In the graph of the $y = \sin x$, let $A \equiv (\alpha, \sin \alpha)$, $B = \left(\frac{\pi}{6}, \sin \frac{\pi}{6}\right)$



clearly slope of OA > slope of OB, so

$$\frac{\sin \alpha}{\alpha} > \frac{\sin \frac{\pi}{6}}{\frac{\pi}{6}} = \frac{3}{\pi} \Rightarrow \frac{\alpha}{\sin \alpha} < \frac{\pi}{3}$$

39. In set P,
 $\sin \theta = (\sqrt{2} + 1) \cos \theta \Rightarrow \tan \theta = \sqrt{2} + 1$
 In set Q, $(\sqrt{2} - 1) \sin \theta = \cos \theta$
 $\Rightarrow \tan \theta = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1 \Rightarrow P = Q$

40. $\cot \alpha_1 \cdot \cot \alpha_2 \dots \cot \alpha_n = 1$
 $\Rightarrow \cos \alpha_1 \cdot \cos \alpha_2 \dots \cos \alpha_n$
 $= \sin \alpha_1 \cdot \sin \alpha_2 \dots \sin \alpha_n$
 Multiplying both sides by
 $\cos \alpha_1 \cdot \cos \alpha_2 \dots \cos \alpha_n$
 $(\cos \alpha_1 \cdot \cos \alpha_2 \dots \cos \alpha_n)^2$
 $= \frac{1}{2^n} \sin 2\alpha_1 \cdot \sin 2\alpha_2 \cdot \sin 2\alpha_n$

$$\leq \frac{1}{2^n} \Rightarrow \cos \alpha_1 \cdot \cos \alpha_2 \dots \cos \alpha_n \leq \frac{1}{2^{n/2}}$$

41. $(a^2 + 4)(b^2 - 1)^2$
 $= (\tan \theta + \cot \theta)^2 (-2 \sin \theta \cos \theta)^2$
 $= 4 \operatorname{cosec}^2 2\theta \cdot \sin^2 2\theta$
42. For maximum value
 $2a^2 - 1 - \cos^2 x = 0$
 $\therefore \cos^2 x = 2a^2 - 1$
 $104 \Rightarrow \sin^2 x = 1 - \cos^2 x = (2 - 2a^2)$
 $\Rightarrow 2a^2 + \sin^2 x = 2$
 $\therefore$ Maximum value of
 $\left| \sqrt{(\sin^2 x + 2a^2)} - \sqrt{(2a^2 - 1 - \cos^2 x)} \right|$
 $= |\sqrt{2} - 0| = \sqrt{2}$

MULTIPLE ANSWER QUESTIONS

43. Given expression is $1 - \cos 2x + \sin 2x = 2k$
 $\Rightarrow \sin 2x - \cos 2x = 2k - 1$
 $\Rightarrow \sin\left(2x - \frac{\pi}{4}\right) = \frac{2k - 1}{\sqrt{2}}$
 $\Rightarrow -1 \leq \frac{2k - 1}{\sqrt{2}} \leq 1$
 $\Rightarrow \frac{1 - \sqrt{2}}{2} \leq k \leq \frac{1 + \sqrt{2}}{2}$
44. $\cos x + \cos y = -\cos z$ (1)
 $\sin x + \sin y = -\sin z$ (2)
 squaring and adding
 $\cos^2 x + \cos^2 y + 2 \cos x \cos y = \cos^2 z$
 $\sin^2 x + \sin^2 y + 2 \sin x \sin y = \sin^2 z$
 $1 + 1 + 2 \cos(x - y) = 1$
 $2(1 + \cos(x - y)) = 1$
 $2 \times 2 \cos^2\left(\frac{x - y}{2}\right) = 1$
 $\cos\left(\frac{x - y}{2}\right) = \pm \frac{1}{2}$
45. $2x = \cos(\alpha - \beta - \gamma + \delta) - \cos(\alpha - \beta + \gamma - \delta)$

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$$2y = \cos(\beta - \gamma - \alpha + \delta) - \cos(\beta - \gamma + \alpha - \delta)$$

and similarly for $2z$

$$\text{Adding } 2x + 2y + 2z = 0$$

$$\Rightarrow x + y + z = 0 \text{ then } x^3 + y^3 + z^3 = 3xyz$$

Hence, a and d are correct answer

$$46. \text{ Given, } \frac{\tan 3A}{\tan A} = k$$

$$\Rightarrow \frac{\tan 3A - \tan A}{\tan A} = k - 1$$

$$\Rightarrow \frac{\sin 2A}{\cos 3A \sin A} = k - 1$$

$$\Rightarrow \frac{2 \cos A}{\cos 3A} = k - 1$$

$$\Rightarrow \frac{\cos A}{\cos 3A} = \frac{k - 1}{2}$$

$\Rightarrow (a)$ is incorrect.

$$\text{Again, } \frac{\tan 3A}{\tan A} = k$$

$$\Rightarrow \frac{\sin 3A}{\cos 3A} \cdot \frac{\cos A}{\sin A} = k$$

$$\Rightarrow \frac{\sin 3A}{\sin A} = k \cdot \frac{2}{k - 1} = \frac{2k}{k - 1}$$

$\Rightarrow (b)$ is correct.

$$\Rightarrow \frac{3 \sin A - 4 \sin^3 A}{\sin A} = \frac{2k}{k - 1}$$

$$\Rightarrow 3 - 4 \sin^2 A = \frac{2k}{k - 1} \text{ or } 4 \sin^2 A = \frac{k - 3}{k - 1}$$

$\Rightarrow$

$$0 < \frac{k - 3}{k - 1} < 4 [\sin A \neq 0 \text{ or } 1]$$

$$\text{Now, } \frac{k - 3}{k - 1} > 0$$

$$\therefore k \in (-\infty, 1) \cup (3, \infty)$$

$$\text{and } \frac{k - 3}{k - 1} < 4$$

$$\Rightarrow \frac{3k - 1}{k - 1} > 0$$

$$\therefore k \in (-\infty, 1/3) \cup (1, \infty)$$

From Eqs. (ii) and (iii), we get

$$k < \frac{1}{3} \text{ and } k > 3.$$

47. From the given conditions

$$2(1 - \sin^2 A) = 3(1 - \sin^2 B) = 3(1 - \sin A)$$

$$\Rightarrow 2 \sin^2 A - 3 \sin A + 1 = 0$$

$$\Rightarrow (2 \sin A - 1)(\sin A - 1) = 0$$

$$\Rightarrow \sin A = 1 \text{ or } \sin A = 1/2$$

$$\Rightarrow A = \pi/2 \text{ or } \pi/6$$

But since a is acute, we have $A = \pi/6$

$$\Rightarrow \sin^2 B = \sin(\pi/6) = 1/2$$

$$\Rightarrow \sin B = 1/\sqrt{2} \Rightarrow B = \pi/4$$

$$48. x = \sum_{n=0}^{\infty} \cos^{2n} \phi = \sec^2 \phi$$

$$y = \sum_{n=0}^{\infty} \sin^{2n} \phi = \sec^2 \phi$$

$$z = \sum_{n=0}^{\infty} \cos^{2n} \phi \sin^{2n} \phi = \frac{1}{1 - \cos^2 \phi \sin^2 \phi}$$

$$= \frac{1}{1 - \frac{1}{xy}} = \frac{xy}{xy - 1}$$

so $xyz = xy + z$ or $xyz = x + y + z$ as $xy = x + y$.

49.

$$f(0) = 0 \Rightarrow a_1 \cos \alpha_1 + a_2 \cos \alpha_2 + \dots + a_n \cos \alpha_n = 0$$

$$f(x_1) = 0 \Rightarrow (a_1 \cos \alpha_1 + a_2 \cos \alpha_2 + \dots + a_n \cos \alpha_n) \cos x_1$$

$$+ (a_1 \sin \alpha_1 + a_2 \sin \alpha_2 + \dots + a_n \sin \alpha_n) \sin x_1 = 0$$

$$\Rightarrow a_1 \sin \alpha_1 + a_2 \sin \alpha_2 + \dots + a_n \sin \alpha_n = 0$$

$$\therefore (x_1 \neq n\pi)$$

$$\therefore a_1 \cos \alpha_1 + a_2 \cos \alpha_2 + \dots + a_n \cos \alpha_n = 0$$

$$\& a_1 \sin \alpha_1 + a_2 \sin \alpha_2 + \dots + a_n \sin \alpha_n = 0$$

$$\Rightarrow f(x) = 0 \forall x$$

50. $\tan \alpha \tan 2\alpha \tan 3\alpha = \tan 3\alpha - \tan 2\alpha - \tan \alpha$
always holds good.

$$\begin{aligned} \text{(B). R.H.S.} &= \frac{\sin 4\alpha + \sin 2\alpha}{\sin 2\alpha \sin 4\alpha} \\ &= \frac{2 \sin 3\alpha \cos \alpha}{\sin 2\alpha \sin 4\alpha} \end{aligned}$$

$$= \frac{1}{\sin \alpha} = \operatorname{cosec} \alpha \quad (\text{using } \sin \pi/7 = \alpha)$$

$\Rightarrow$ (b) is correct

C. $\cos \alpha + \cos 3\alpha + \cos 5\alpha$

$$= \frac{\sin 3\alpha}{\sin \alpha} \cos 3\alpha = \frac{\sin 6\alpha}{2 \sin \alpha} = \frac{\sin \frac{6\pi}{7}}{2 \sin \frac{\pi}{7}}$$

$$= \frac{\sin \left(\pi + \frac{\pi}{7} \right)}{2 \sin \frac{\pi}{7}} = \frac{1}{2}$$

$$\text{D. } 8 \cos \alpha \cos 2\alpha \cos 4\alpha = \frac{\sin 8\alpha}{\sin \alpha}$$

$$= \frac{\sin \frac{8\pi}{7}}{\sin \frac{\pi}{7}} = -\frac{\sin \frac{\pi}{7}}{\sin \frac{\pi}{7}} = -1$$

COMPREHENSION TYPE

51. Conceptual

$$\begin{aligned} 52. \quad & \sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14} = \sin \frac{5\pi}{14} \sin \frac{3\pi}{14} \sin \frac{\pi}{14} \\ &= \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} = -\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \\ &= -\frac{\sin \frac{8\pi}{7}}{8 \sin \frac{\pi}{7}} = \frac{1}{8} \end{aligned}$$

Hence, (c) is the correct answer

$$\begin{aligned} 53. \quad & \cos 2^3 \frac{\pi}{10} \cos 2^4 \frac{\pi}{10} \cos 2^5 \frac{\pi}{10} \dots \cos 2^{10} \frac{\pi}{10} \\ &= \frac{\sin 2^{11} \frac{\pi}{10}}{256 \sin 2^3 \frac{\pi}{10}} = \frac{1}{256} \end{aligned}$$

$$\begin{aligned} 54. \quad & \cos \frac{\pi}{11} \cos \frac{2\pi}{11} \cos \frac{3\pi}{11} \dots \cos \frac{10\pi}{11} \\ &= \left(\cos \frac{\pi}{11} \cos \frac{2\pi}{11} \cos \frac{3\pi}{11} \cos \frac{4\pi}{11} \cos \frac{5\pi}{11} \right)^2 \\ &= \left(\cos \frac{\pi}{11} \cos \frac{2\pi}{11} \cos \frac{4\pi}{11} \cos \frac{8\pi}{11} \cos \frac{5\pi}{11} \right)^2 \\ &= \left(\frac{\sin 16 \frac{\pi}{11}}{16 \sin \frac{\pi}{11}} \cdot \cos \frac{5\pi}{11} \right)^2 = \left(\frac{2 \sin \frac{5\pi}{11} \cos \frac{5\pi}{11}}{32 \sin \frac{\pi}{11}} \right)^2 = \frac{1}{1024} \end{aligned}$$

55. We have, $\cos A \cos B \cos C = \frac{\sqrt{3}-1}{8}$ and

$$\sin A \sin B \sin C = \frac{3+\sqrt{3}}{8}$$

$$\Rightarrow \tan A \tan B \tan C = \frac{3+\sqrt{3}}{\sqrt{3}-1} \quad \dots\dots\dots(i)$$

But, $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$
(using conditional identity)

$$\tan A + \tan B + \tan C = \frac{3+\sqrt{3}}{\sqrt{3}-1} \quad \dots\dots\dots(ii)$$

56. Now, $A + B + C = \pi \Rightarrow \cos(A + B + C) = \cos \pi \Rightarrow \cos A \cos B \cos C \{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A\} = -1$

$$\Rightarrow \frac{\sqrt{3}-1}{8}$$

$$\begin{aligned} \{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A\} &= -1 \\ \Rightarrow \tan A \tan B + \tan B \tan C + \tan C \tan A &= 5 + 4\sqrt{3} \end{aligned}$$

57. If $A = 45^\circ$, $B = 60^\circ$, then $\tan C$

$$\begin{aligned} &= 2 + \sqrt{3} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = \frac{1+(1/\sqrt{3})}{1-(1/\sqrt{3})} \\ &= \tan(45^\circ + 30^\circ) \\ &\Rightarrow C = 75^\circ. \end{aligned}$$

58. (i) $\Rightarrow (x + A)(8x^3 - 4x^2 - 4x + A) = 0 \quad \dots\dots\dots(i)$

JEE MAINS ADVANCED

But $x + 1 = 0 \Rightarrow \cos \theta = x = -1 \Rightarrow x = \cos \pi$
 $= \cos\left(\frac{7\pi}{7}\right) \therefore$ the roots of $8x^3 - 4x^2 - 4x + 1$

$= 0$, are $\cos \frac{\pi}{7}$, $\cos \frac{3\pi}{7}$, $\cos \frac{5\pi}{7}$ (ii)

59. Put, $x = 1/y$ in (ii) i.e., $(y = \sec \theta)$

then $\sec \frac{\pi}{7}$, $\sec \frac{3\pi}{7}$, $\sec \frac{5\pi}{7}$ are roots of

$$a \frac{8}{y^3} - \frac{4}{y^2} - \frac{4}{y} + 1 = 0$$

$$\Rightarrow y^3 - 4y^2 - 4y + 8 = 0$$

$\therefore \sec \frac{\pi}{7} + \sec \frac{3\pi}{7} + \sec \frac{5\pi}{7}$ is 4(iii)

60. (a) and (b) are clearly false, since $f(x)$ is not defined for $\frac{k\pi}{2}$, On the contrary the two RHS included in choices (a) and (b) are defined for all x .

(c) is true since $f(x) = \sin x \cos x - \cos x \sin x$

61. If $x \neq \frac{n\pi}{2}$, then

$$f(x) = \sin x |\cos x| + \cos x |\sin x|$$

$\Rightarrow f(x) = 0$, if $\sin x$ and $\cos x$ have opposite signs

$$\Rightarrow \sin x \cos x < 0$$

$$\Rightarrow \sin 2x < 0$$

$$\Rightarrow 2n\pi + \pi < 2x < 2n\pi + 2\pi \text{ or}$$

$$(2n\pi + 1)\frac{\pi}{2} < x < (2n + 2)\frac{\pi}{2}$$

$\Rightarrow$ (b) is correct

62. For $x \neq \frac{n\pi}{2}$

$$f(x) = \sin x |\cos x| + \cos x |\sin x|$$

Let us solve $\sin x |\cos x| + \cos x |\sin x| < 0$

If $\sin x$ and $\cos x$ both are positive, then this is not satisfied. If they have opposite signs then the inequality cannot be satisfied either ($\because LHS = 0$). If both $\sin x$ and $\cos x$ are negative, then it is satisfied for all x . Thus, $f(x) < 0$ if,

$$\sin x < 0, \cos x < 0 \text{ and } x \neq \frac{n\pi}{2}$$

$$\Rightarrow 2n\pi + \pi < x < 2n\pi + \pi + \frac{\pi}{2}$$

$$\Rightarrow (2n+1)\pi < x < (4n+3)\frac{\pi}{2} \Rightarrow \text{(c) is correct}$$

MATCHING TYPE

63. A) $\cos(2A + \theta) + \cos(2B + \theta)$

$$= 2 \cos(A + B + \theta) \cos(A - B)$$

Maximum value is $2\cos(A-B)$ when

$$\cos(A + B + \theta) = 1$$

$$B) \{\cos 2A + \cos 2B\}$$

$$2\cos(A+B)\cos(A-B)$$

Maximum value is $2\cos(A-B)$ when $\cos(A+B)=1$

C) For $y = \sec x$, $x \in (0, \pi/2)$, tangent drawn to it any point lies completely below the graph of

$$y = \sec x, \text{ thus } \frac{\sec 2A + \sec 2B}{2} \geq \sec(A + B)$$

$$\Rightarrow \sec 2A + \sec 2B \geq 2\sec(A + B)$$

Hence minimum value is $A+B$

$$D) \sqrt{\{\tan \theta + \cot \theta - 2\cos 2(A + B)\}}$$

$$= \sqrt{(\sqrt{\tan \theta} - \sqrt{\cot \theta})^2 + 2 - 2\cos 2(A + B)}$$

$$= \sqrt{(\sqrt{\tan \theta} - \sqrt{\cot \theta})^2 + 4\sin^2(A + B)}$$

Minimum value occurs when $\sqrt{\tan \theta} = \sqrt{\cot \theta}$ and Minimum value is

$$\sqrt{4\sin^2(A + B)} = 2\sin(A + B)$$

64. Let $y = \frac{7 + 6\tan \theta - \tan^2 \theta}{(1 + \tan^2 \theta)}$

$$= 7\cos^2 \theta + 6\sin \theta \cos \theta - \sin^2 \theta$$

$$= 7\left(\frac{1 + \cos 2\theta}{2}\right) + 3\sin 2\theta - \left(\frac{1 - \cos 2\theta}{2}\right)$$

$$= 3\sin 2\theta + 4\cos 2\theta + 3$$

$$-\sqrt{(3^2 + 4^2)} + 3 \leq 3\sin 2\theta + 4\cos 2\theta + 3$$

$$\leq \sqrt{(3^2 + 4^2)} + 3$$

$$\therefore -2 \leq y \leq 8$$

$$\Rightarrow \lambda = 8, \mu = -2$$

$$\Rightarrow \lambda + \mu = 6, \lambda - \mu = 10(r, s)$$

65. Conceptual

ASSERTION-REASONING TYPE

66. The minimum value of sum of three sines can be -3 provided it is attainable through some angle. Indeed value -3 is possible if $\sin \alpha = -1, \sin \beta = -1, \sin \gamma = -1$

$$\Rightarrow \alpha = (4l-1)\frac{\pi}{2}, \beta = (4m-1)\frac{\pi}{2}, \gamma = (4n-1)\frac{\pi}{2}$$

$$\text{Now } \alpha + \beta + \gamma = \pi \Rightarrow 4(l+m+n) = 5$$

Since l, m, n are integers $\Rightarrow$ minimum cannot be -3

67. Conceptual

68. Let $g(x) = 3 \sin x + 4 \cos x - 2$

$$\text{Maximum value of } g(x) = \sqrt{3^2 + 4^2} - 2 = 3$$

$$\text{Minimum value of } g(x) = -\sqrt{3^2 + 4^2} - 2 = -7$$

Therefore the range of $f(x) = \frac{1}{g(x)}$ is

$$R - \left[-\frac{1}{7}, \frac{1}{3} \right]$$

$\Rightarrow$ it is an unbounded function

$\Rightarrow f(x)$ has no maximum and no minimum values

INTEGER TYPE QUESTIONS

69. $\tan 3x = 0 \Rightarrow x = \frac{n\pi}{3}, n \in \mathbb{Z}$

70.
$$\begin{aligned} \text{LHS} &= 16 \left(\cos \theta - \cos \frac{\pi}{8} \right) \left(\cos \theta - \cos \frac{3\pi}{8} \right) \\ &\times \left(\cos \theta - \cos \frac{5\pi}{8} \right) \left(\cos \theta - \cos \frac{7\pi}{8} \right) \\ &= 16 \left(\cos^2 \theta - \cos^2 \frac{\pi}{8} \right) \left(\cos^2 \theta - \cos^2 \frac{3\pi}{8} \right) \\ &= 16 \left(\cos^4 \theta - \cos^2 \theta + \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \right) \\ &= 16 \left(\cos^4 \theta - \cos^2 \theta + \frac{1}{8} \right) \\ &= 16 \left(-\cos^2 \theta \sin^2 \theta + \frac{1}{8} \right) = 16 \left(\frac{-\sin^2 2\theta}{4} + \frac{1}{8} \right) \\ &= 16 \left(\frac{1-2\sin^2 2\theta}{8} \right) = 16 \left(\frac{\cos^2 2\theta - \sin^2 2\theta}{8} \right) \\ &= \frac{16 \cos 4\theta}{8} \end{aligned}$$

71.
$$\begin{aligned} f(n\theta) &= \frac{2 \sin 2\theta}{\cos 2\theta - \cos 4n\theta} \\ &= \frac{2 \sin 2\theta}{2 \sin(2n+1)\theta \sin(2n-1)\theta} \\ &= \frac{\sin((2n+1)\theta - (2n-1)\theta)}{\sin(2n+1)\theta \sin(2n-1)\theta} \\ &= \frac{\sin(2n+1)\theta \cos(2n-1)\theta - \cos(2n+1)\theta \sin(2n-1)\theta}{\sin(2n+1)\theta \sin(2n-1)\theta} \end{aligned}$$

$$= \cot(2n-1)\theta - \cot(2n+1)\theta$$

72. $\therefore$

$$(\cos 27^\circ + \sin 27^\circ)^2 = 1 + \sin 54^\circ = 1 + \cos 36^\circ$$

$$\Rightarrow \cos 27^\circ + \sin 27^\circ = \sqrt{1 + \cos 36^\circ}$$

$$\text{Also, } \cos 27^\circ - \sin 27^\circ = \sqrt{1 - \cos 36^\circ}$$

$$(\because \cos 27^\circ > \sin 27^\circ) \quad 104.$$

$$\therefore 2 \sin 27^\circ = \sqrt{1 + \cos 36^\circ} - \sqrt{1 - \cos 36^\circ}$$

$$= \sqrt{1 + \left(\frac{\sqrt{5}+1}{4} \right)} - \sqrt{1 - \left(\frac{\sqrt{5}+1}{4} \right)}$$

$$\therefore 4 \sin 27^\circ = \sqrt{5 + \sqrt{5}} - \sqrt{3 - \sqrt{5}}$$

on comparing, we get

$$\alpha = 5 + \sqrt{5}, \beta = 3 - \sqrt{5}$$

$$\therefore \alpha + \beta = 8, \alpha\beta = 10 - 2\sqrt{5}$$

$$\alpha + \beta - \alpha\beta + 2 = 2\sqrt{5}$$

$$\therefore (\alpha + \beta - \alpha\beta + 2)^4 = 400$$

73.
$$\therefore \prod_{r=1}^4 \sin(rA) = \sin A \sin 2A \sin 3A \sin 4A$$

$$= \sin A \cdot 2 \sin A \cos A \cdot (3 \sin A$$

$$- 4 \sin^3 A \cdot 2 \sin 2A \cos 2A)$$

$$= 2 \sin^2 A \cos A \cdot \sin A (3 - 4 \sin^2 A)$$

$$\cdot 4 \sin A \cos A (1 - 2 \sin^2 A)$$

$$= 8x^2(1-x)(3-4x)(1-2x)$$

$$= 24x^2 - 104x^3 + 144x^4 - 64x^5$$

on comparing, we get $a=24, b=-104, c=144,$

$$d = -64$$

$$74. \therefore \cot(\theta - \alpha), 3 \cot \theta, \cot(\theta + \alpha) \text{ are in AP}$$

$$6 \cot \theta = \cot(\theta - \alpha) + \cot(\theta + \alpha)$$

$$\frac{6 \cos \theta}{\sin \theta} = \frac{\sin 2\theta}{\sin(\theta + \alpha) \sin(\theta - \alpha)}$$

$$\Rightarrow 6 \cos \theta \{\sin^2 \theta - \sin^2 \alpha\} = 2 \sin^2 \theta \cos \theta$$

$$\Rightarrow 3(\sin^2 \theta - \sin^2 \alpha) = \sin^2 \theta$$

$$\therefore 2 \sin^2 \theta = 3 \sin^2 \alpha$$

$$\text{or } \frac{2 \sin^2 \theta}{3 \sin^2 \alpha} = 1$$

$$\therefore \frac{4 \sin^2 \theta}{3 \sin^2 \alpha} = 2 \times \frac{2 \sin^2 \theta}{3 \sin^2 \alpha} = 2$$

$$75. \therefore \sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

$$= 32 \sin A/2 \sin B/2 \sin C/2$$

$$\cos A/2 \cos B/2 \cos C/2 \text{ and}$$

$$\sin A + \sin B + \sin C = 4 \cos A/2 \cos B/2 \cos C/2$$

(from conditional identities)

$$\therefore \frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C}$$

$$= 8 \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right)$$

$$\text{on comparing, we get } \lambda = 8$$

$$76. \text{ We have}$$

$$\left(a \tan \beta - \sqrt{a^2 - 1} \tan \alpha\right)^2$$

$$+ \left(\sqrt{a^2 + 1} \tan \beta - \sqrt{a^2 - 1} \tan \gamma\right)^2$$

$$+ \left(a \tan \gamma - \sqrt{a^2 + 1} \tan \alpha\right)^2 \geq 0$$

$$\Rightarrow \{a^2 + a^2 - 1 + a^2 + 1\}(\tan^2 \alpha + \tan^2 \beta + \tan^2 \gamma)$$

$$- \{a \tan \alpha + \sqrt{a^2 - 1} \tan \beta + \sqrt{a^2 + 1} \tan \gamma\}^2 \geq 0$$

$$\Rightarrow \tan^2 \alpha + \tan^2 \beta + \tan^2 \gamma \geq \frac{4}{3}$$

$$77. \frac{1}{4 \cos^2 \theta + 1 + \frac{3}{2} \sin 2\theta}$$

$$\Rightarrow \frac{1}{2[1 + \cos 2\theta] + 1 + \frac{3}{2} \sin 2\theta}$$

$$\text{lies between } \frac{1}{2} \text{ to } \frac{11}{2}$$

$$\therefore \text{ maximum value is } 2.$$

$$\text{Minimum value of } 1 + 4 \cos^2 \theta + 3 \sin \theta \cos \theta$$

$$1 + \frac{4(1 + \cos 2\theta)}{2} + \frac{3}{2} \sin 2\theta$$

$$= 1 + 2 + 2 \cos 2\theta + \frac{3}{2} \sin 2\theta$$

$$3 + 2 \cos 2\theta + \frac{3}{2} \sin 2\theta$$

$$\therefore = 3 - \sqrt{4 + \frac{9}{4}} = 3 - \frac{5}{2} = \frac{1}{2}$$

$$\text{So maximum value of}$$

$$\frac{1}{4 \cos^2 \theta + 1 + \frac{3}{2} \sin 2\theta} \text{ is } 2$$

$$78. \frac{1}{\sin \frac{\pi}{n}} - \frac{1}{\sin \frac{3\pi}{n}} = \frac{1}{\sin \frac{2\pi}{n}} \Rightarrow \frac{\sin \frac{3\pi}{n} - \sin \frac{\pi}{n}}{\sin \frac{\pi}{n} \sin \frac{3\pi}{n}}$$

$$= \frac{1}{\sin \frac{2\pi}{n}} \frac{\left(2 \sin \frac{\pi}{n} \cos \frac{2\pi}{n}\right) \sin \frac{2\pi}{n}}{\sin \frac{\pi}{n} \sin \frac{3\pi}{n}} = 1$$

$$\Rightarrow \sin \frac{4\pi}{n} = \sin \frac{3\pi}{n} \Rightarrow \frac{4\pi}{n} + \frac{3\pi}{n} = \pi \Rightarrow n = 7$$

SUBJECTIVE TYPE QUESTIONS

$$79. \sin \theta + \cos \theta \leq \sqrt{2} < \frac{\pi}{2}$$

$$\Rightarrow \sin \theta < \frac{\pi}{2} - \cos \theta$$

$$\Rightarrow \cos(\sin \theta) > \cos\left(\frac{\pi}{2} - \cos \theta\right) = \sin(\cos \theta)$$

$$80. e^{-\frac{\pi}{2}} < \theta < e^{\frac{\pi}{2}} \Rightarrow$$

$$-\frac{\pi}{2} < \ln \theta < \frac{\pi}{2}$$

$$\therefore \cos(\ln \theta) > 0$$

$$\cos \theta < 1 \Rightarrow \ln(\cos \theta) < 0$$

$$\therefore \cos(\ln \theta) \text{ is the larger}$$

$$81. \sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2n-1)\theta$$

$$= \frac{\sin \left[n \left(\frac{2\theta}{2} \right) \right]}{\sin \left(\frac{2\theta}{2} \right)} \sin \left(\frac{\theta + (2n-1)\theta}{2} \right) = \frac{\sin^2 n\theta}{\sin \theta}$$

$$82. \text{ Conceptual}$$

$$83. \text{ Conceptual}$$

$$84. \text{ Let } \theta = \frac{\pi}{7} \Rightarrow 3\theta = \pi - 4\theta$$

$$\Rightarrow \sin 3\theta = \sin 4\theta$$

$$\Rightarrow 3\sin \theta - 4\sin^3 \theta = 4\sin \theta \cos \theta \cos 2\theta$$

$$\Rightarrow 3 - 4\sin^2 \theta = 4\cos \theta (2\cos^2 \theta - 1)$$

$$\Rightarrow 8\cos^3 \theta - 4\cos \theta = 4\cos^2 \theta - 1$$

$$\Rightarrow 4\cos \theta (2\cos^2 \theta - 1) = 4\cos^2 \theta - 1$$

$$\Rightarrow 4\cos \theta = \frac{4\cos^2 \theta - 1}{2\cos^2 \theta - 1} = \frac{3 - \tan^2 \theta}{1 - \tan^2 \theta}$$

$$85. \text{ Conceptual}$$

$$86.$$

$$\frac{\sin 1^\circ}{\sin X^\circ \sin(X+1)^\circ} = \cot X^\circ - \cot(X+1)^\circ = \cot 45^\circ - \cot 46^\circ + \dots$$

$$= \cot 45^\circ - (\cot 46^\circ + \cot 134^\circ) + (\cot 47^\circ + \cot 133^\circ) -$$

$$\dots - (\dots) - \cot 90^\circ$$

$$= 1$$

$$87. \text{ Here, } \cot \theta = \cot A + \cot B + \cot C$$

$$\Rightarrow \cot(\theta) - \cot(A) = \cot B + \cot C$$

$$\Rightarrow \frac{\cos \theta}{\sin \theta} - \frac{\cos A}{\sin A} = \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C}$$

$$\Rightarrow$$

$$\frac{\cos \theta \sin A - \cos A \sin \theta}{\sin \theta \sin A} = \frac{\cos B \sin C + \sin B \cos C}{\sin B \sin C}$$

$$\Rightarrow \frac{\sin(A - \theta)}{\sin A \sin \theta} = \frac{\sin(B + C)}{\sin B \sin C}$$

$$\Rightarrow \sin(A - \theta) = \frac{\sin^2 A \sin \theta}{\sin B \sin C} \dots (i)$$

$$\text{Similarly, } \sin(B - \theta) = \frac{\sin^2 B \sin \theta}{\sin A \sin C} \dots (ii)$$

$$\text{and } \sin(C - \theta) = \frac{\sin^2 C \sin \theta}{\sin A \sin B} \dots (iii)$$

Multiplying (i), (ii) and (iii), we get,

$$\sin(A - \theta) \cdot \sin(B - \theta) \cdot \sin(C - \theta) = \sin^3 \theta.$$

$$88. \cos B + \cot C = -\cot A$$

$$\Rightarrow \frac{\sin(B + C)}{\sin B \sin C} = \frac{-\cos A}{\sin A}$$

$$\frac{\sin^2 A}{\sin B \sin C} = -\cos A$$

$$89. \frac{1}{\sin \theta} = \frac{\sin \theta / 2}{S\theta S\theta / 2} = \frac{S(\theta - \theta / 2)}{\sin \theta \sin \theta / 2}$$

$$= \cot \theta / 2 - \cot \theta$$

$$90. \text{ Let } \theta = \frac{n\pi}{4} (n = 1, 2, \dots, 7) 4\theta = n\pi - 3\theta$$

$$\tan 4\theta = -\tan 3\theta \Rightarrow t^{-7} - 21t^5 + 35t^3 - 7t = 0$$

$$t = \tan \left(\frac{n\pi}{7} \right), n = 1, 2, 3, 4, 5, 6, 7$$

$$t^6 - 21t^4 + 35t^2 - 7 = 0, \quad t^2 = X (t \neq 0)$$

$$X^3 - 21X^2 + 35X - 7 = 0$$

$$\text{Roots are } \tan^2 \frac{\pi}{7}, \tan^2 \frac{2\pi}{7}, \tan^2 \frac{3\pi}{7}$$

$$\tan^2 \frac{5\pi}{7} = \tan^2 \frac{2\pi}{7}, \tan^2 \frac{4\pi}{7} = \tan^2 \frac{3\pi}{7},$$

$$\tan^2 \frac{6\pi}{7} = \tan^2 \frac{\pi}{7}$$

$$\text{sum} = 21$$

$$91. \frac{X - Y}{1 + X.Y} + \frac{Y - Z}{1 + Y.Z} + \frac{Z - X}{1 + Z.X} = 0$$

$$\Rightarrow (X - Y)(Y - Z)(Z - X) = 0$$

$$92. \cos \alpha = t \cos \beta \Rightarrow \sin \alpha = -(1 + t) \sin \beta$$

$$\Rightarrow 1 - t^2 \cos^2 \beta = (1+t)^2 \sin^2 \beta$$

$$\Rightarrow \sin^2 \beta = \frac{1-t^2}{2t+1}$$

$$\Rightarrow \cos^2 \beta = \frac{2t+t^2}{1+2t}$$

93. $\tan r\alpha \cdot \tan(r+1)\alpha + 1$

$$= \frac{\sin r\alpha \cdot \sin(r+1)\alpha + \cos(r\alpha) \cdot \cos(r+1)\alpha}{\cos r\alpha \cdot \cos(r+1)\alpha}$$

$$= \frac{\cos \alpha}{\cos r\alpha \cdot \cos(r+1)\alpha}$$

$$= \cot \alpha \cdot \frac{\sin \alpha}{\cos r\alpha \cdot \cos(r+1)\alpha}$$

$$= \cot \alpha \cdot \frac{\sin \{(r+1)\alpha - r\alpha\}}{\cos r\alpha \cdot \cos(r+1)\alpha}$$

$$= \cot \alpha \cdot \frac{\sin(r+1)\alpha \cdot \cos(r\alpha) - \cos(r+1)\alpha \cdot \sin(r\alpha)}{\cos r\alpha \cdot \cos(r+1)\alpha}$$

$$= \cot \alpha \cdot \{\tan(r+1)\alpha - \tan(r)\alpha\}$$

$$\therefore \sum_{r=1}^n \tan r\alpha \tan(r+1)\alpha + \sum_{r=1}^n 1$$

$$= \cot \alpha \sum_{r=1}^n \{\tan(r+1)\alpha - \tan r\alpha\}$$

$$\text{or } \sum_{r=1}^n \tan r\alpha \cdot \tan(r+1)\alpha + n$$

$$= \cot \alpha \left\{ (\tan 2\alpha - \tan \alpha) + (\tan 3\alpha - \tan 2\alpha) + (\tan 4\alpha - \tan 3\alpha) + \dots + \right.$$

$$\left. (\tan(n+1)\alpha - \tan n\alpha) \right\}$$

$\therefore$

$$L.H.S. = -n + \cot \alpha \{\tan(n+1)\alpha - \tan \alpha\}$$

$$= -n + \cot \alpha \cdot \tan(n+1)\alpha - \cot \alpha \cdot \tan \alpha$$

$$= \cot \alpha \cdot \tan(n+1)\alpha - n - 1 = R.H.S.$$

94. Square - ADD

$$\sin(A+B) = \frac{1}{2}$$

$$\therefore A+B = 150^\circ \Rightarrow \angle C = 30^\circ$$

$$A+B = 30^\circ \Rightarrow \angle C = 150^\circ \Rightarrow A < 30^\circ$$

$$(\text{or}) \Rightarrow 3 \sin A < 3/2$$

$$\Rightarrow 3 \sin A + 4 \cos B < \frac{3}{2} + 4.1 < 6$$

It is contradiction

95. $a = \sqrt{2} \cos \theta, b = \sqrt{2} \sin \theta$

$$\alpha = \sqrt{2} \cos \phi, \beta = \sqrt{2} \sin \phi$$

$$\Rightarrow S = 2 - 2[\sin(\theta + \pi/4) + \sin(\phi + \pi/4)]$$

$$+ 2[\cos(\theta - \phi)]$$

$$= 2+4+2=8$$

LEVEL - VI

SINGLE ANSWER QUESTIONS

1. If $\cos(A+B+C) = \cos A \cos B \cos C$, then

$$\frac{8 \sin(B+C) \sin(C+A) \sin(A+B)}{\sin 2A \sin 2B \sin 2C} =$$

A) 1 B) -1 C) $\frac{1}{2}$ D) $-\frac{1}{2}$

2. $\sin^4 \frac{\pi}{16} + \sin^4 \frac{3\pi}{16} + \sin^4 \frac{5\pi}{16} + \sin^4 \frac{7\pi}{16} =$

A) 1 B) $1/2$ C) $3/2$ D) 2

3. $a^2 + b^2 + 2ab \cos \theta = 1,$

$$c^2 + d^2 + 2cd \cos \theta = 1 \text{ and}$$

$$ac + bd + (ad + bc) \cos \theta = 0 \text{ then } a^2 + c^2 =$$

A) $\sec^2 \theta$ B) $\operatorname{cosec}^2 \theta$

C) $\cot^2 \theta$ D) $\tan^2 \theta$

4. If $a \sin x + b \cos(x + \theta) + b \cos(x - \theta) = d$ then the minimum value of $|\cos \theta|$ is equal to (where x is variable)

A) $\frac{1}{2|b|} \sqrt{d^2 - a^2}$ B) $\frac{1}{2|a|} \sqrt{d^2 - a^2}$
C) $\frac{1}{2|d|} \sqrt{d^2 - a^2}$ D) $\frac{1}{|d|} \sqrt{d^2 - a^2}$

5. $\cos^6 \frac{\pi}{16} + \cos^6 \frac{3\pi}{16} + \cos^6 \frac{5\pi}{16} + \cos^6 \frac{7\pi}{16} =$
A) 0 B) $1/4$ C) $3/4$ D) $5/4$

6. If the equation

$\cot^4 x - 2 \cos e c^2 x + a^2 = 0$ has at least one solution, then the sum of all possible integral values of 'a' is equal to

A) 4 B) 3 C) 2 D) 0

7. The value of $\sum_{n=1}^{\infty} \frac{\tan\left(\frac{\theta}{2^n}\right)}{2^{n-1} \cos \frac{\theta}{2^{n-1}}}$ is

A) $\frac{2}{\sin 2\theta} - \frac{1}{\theta}$ B) $\frac{2}{\sin 2\theta} + \frac{1}{\theta}$
C) $\frac{1}{\sin 2\theta} - \frac{1}{\theta}$ D) $\frac{1}{\sin \theta} - \frac{1}{\theta}$

8. The sum of the series,

$\sin \theta \sec 3\theta + \sin 3\theta \sec 3^2 \theta + \sin 3^2 \theta \sec 3^3 \theta + \dots$ upto n terms, is

A) $\frac{1}{2} [\tan 3^n \theta - \tan 3^{n-1} \theta]$
B) $[\tan 3^n \theta - \tan \theta]$
C) $\frac{1}{2} [\tan 3^n \theta - \tan \theta]$
D) $\frac{1}{2} (\tan 3^n \theta - 1)$

9. If $\frac{\sec^8 \theta}{a} + \frac{\tan^8 \theta}{b} = \frac{1}{a+b}$ then

A) $ab \leq 0$ B) $ab \geq 0$ C) $a+b=0$ D) $ab=0$

10. If the mapping $f(x) = ax + b, a < 0$ maps $[-1, 1]$ onto $[0, 2]$ then for all values of $\theta, A = \cos^2 \theta + \sin^4 \theta$ is such that

A) $f\left(\frac{1}{4}\right) \leq A \leq f(0)$ B) $f(0) \leq A \leq f(-1)$
C) $f\left(\frac{1}{3}\right) \leq A \leq f(0)$ D) $f(1) < A \leq f(-1)$

11. If $x \sin a + y \sin 2a + z \sin 3a = \sin 4a$

$x \sin b + y \sin 2b + z \sin 3b = \sin 4b,$

$x \sin c + y \sin 2c + z \sin 3c = \sin 4c.$

Then the roots of the equation

$t^3 - \left(\frac{z}{2}\right)t^2 - \left(\frac{y+z}{4}\right)t + \left(\frac{z-x}{8}\right) = 0, a, b, c \neq n\pi,$ are

A) $\sin a, \sin b, \sin c$
B) $\cos a, \cos b, \cos c$
C) $\sin 2a, \sin 2b, \sin 2c$
D) $\cos 2a, \cos 2b, \cos 2c$

12. If α, β, γ are acute angles and

$\cos \theta = \sin \beta / \sin \alpha, \cos \phi = \sin \gamma / \sin \alpha$ and

$\cos(\theta - \phi) = \sin \beta \sin \gamma$ then the value of

$\tan^2 \alpha - \tan^2 \beta - \tan^2 \gamma$ is equal to

A) -1 B) 0 C) 1 D) 2

13. The value of $\sum_{k=1}^{100} \sin(kx) \cos(101-k)x$ is equal to

(A) $\frac{101}{2} \sin(101x)$ (B) $99 \sin(101x)$
(C) $50 \sin(101x)$ (D) $100 \sin(101x)$

MULTIPLE ANSWERS QUESTIONS

14. $x = \sqrt{a^2 \cos^2 \alpha + b^2 \sin^2 \alpha} + \sqrt{a^2 \sin^2 \alpha + b^2 \cos^2 \alpha}$,
then $x^2 = a^2 + b^2 + 2\sqrt{p(a^2 + b^2) - p^2},$

JEE MAINS ADVANCED

where p is equal to

A) $a^2 \cos^2 \alpha + b^2 \sin^2 \alpha$

B) $a^2 \sin^2 \alpha + b^2 \cos^2 \alpha$

C) $\frac{1}{2} [a^2 + b^2 + (a^2 - b^2) \cos 2\alpha]$

D) $\frac{1}{2} [a^2 + b^2 - (a^2 - b^2) \cos 2\alpha]$

15. If $(x - a) \cos \theta + y \sin \theta = (x - a) \cos \phi + y \sin \phi = a$ and $\tan(\theta/2) - \tan(\phi/2) = 2b$ then

A) $y^2 = 2ax - (1 - b^2)x^2$

B) $\tan \frac{\theta}{2} = \frac{1}{x}(y + bx)$

C) $y^2 = 2bx - (1 - a^2)x^2$

D) $\tan \frac{\phi}{2} = \frac{1}{x}(y - bx)$

16. If $\cos x - \sin \alpha \cot \beta \sin x = \cos \alpha$ then the value of $\tan(x/2)$ is

A) $-\tan(\alpha/2) \cot(\beta/2)$

B) $\tan(\alpha/2) \tan(\beta/2)$

C) $-\cot(\alpha/2) \tan(\beta/2)$

D) $\cot(\alpha/2) \cot(\beta/2)$

17. If $\sin x + \cos x + \tan x + \cot x + \sec x + \operatorname{cosec} x = 7$ and $\sin 2x = a - b\sqrt{7}$ then

A) $a = 8$ B) $b = 22$ C) $a = 22$ D) $b = 8$

18. The expression $(a \tan g + b \cot g)$

$(a \cot g + b \tan g) - 4ab \cot^2 2g$ is

(A) independent of a, b

(B) independent of g (C) dependent on g

(D) dependent on a, b

19. The equation $x^3 - \frac{3}{4}x = -\frac{\sqrt{3}}{8}$ is satisfied by

(A) $x = \cos\left(\frac{5\pi}{18}\right)$ (B) $x = \cos\left(\frac{7\pi}{18}\right)$

(C) $x = \cos\left(\frac{23\pi}{18}\right)$ (D) $x = -\sin\left(\frac{7\pi}{9}\right)$

COMPREHENSION QUESTIONS

- (P) $\alpha, \beta, \gamma, \delta$ are angles in I, II, III and IV quadrant respectively and no one of them is an integral multiple of $\pi/2$. They form an increasing arithmetic progression.

20. Which of the following holds

A) $\cos(\alpha + \delta) > 0$ B) $\cos(\alpha + \delta) = 0$

C) $\cos(\alpha + \delta) < 0$

D) $\cos(\alpha + \delta) > 0$ or $\cos(\alpha + \delta) < 0$

21. Which of the following does not hold

A) $\sin(\beta + \gamma) = \sin(\alpha + \delta)$

B) $\sin(\beta - \gamma) = \sin(\alpha - \delta)$

C) $\tan 2(\alpha - \beta) = \tan(\beta - \delta)$

D) $\cos(\alpha + \gamma) = \cos 2\beta$

22. If $\alpha + \beta + \gamma + \delta = \theta$ and $\alpha = 70^\circ$,

A) $400^\circ < \theta < 580^\circ$ B) $470^\circ < \theta < 650^\circ$

C) $680^\circ < \theta < 860^\circ$ D) $540^\circ < \theta < 900^\circ$

- (P) A line OA of length r starts from its initial position OX and traces an angle $\angle AOB = \alpha$ in the anticlockwise direction. It then traces back in the clockwise direction an angle $\angle BOC = 3\theta$ (where $\alpha > 3\theta$). L is the foot of the perpendicular from C on OA.

$$\frac{\sin^3 \theta}{CL} = \frac{\cos^3 \theta}{OL} = 1$$

23. $\frac{1 - r \cos \alpha}{r \sin \alpha}$ is equal to

A) $\tan 2\theta$ B) $\cot 2\theta$ C) $\sin 2\theta$ D) $\cos 2\theta$

24. $\frac{2r \sin \alpha}{1 + 2r \cos \alpha}$ is equal to

A) $\tan^2 \theta$ B) $\cot^2 \theta$ C) $\cot 2\theta$ D) $\tan 2\theta$

25. $\frac{2r^2 - 1}{r}$ is equal to

A) $\sin \alpha$ B) $\cos \alpha$ C) $\sin \theta$ D) $\cos \theta$

MATRIX MATCHING QUESTIONS

26. Let $f_n(\theta) =$

$$\frac{\cos \frac{\theta}{2} + \cos 2\theta + \cos \frac{7\theta}{2} + \dots + \cos (3n-2)\frac{\theta}{2}}{\sin \frac{\theta}{2} + \sin 2\theta + \sin \frac{7\theta}{2} + \dots + \sin (3n-2)\frac{\theta}{2}}$$

Then match the entries of column-I with their corresponding values given in column-II.

Column-I	Column-II
(A) $f_3 \left(\frac{3\pi}{16} \right)$	(P) $2 - \sqrt{3}$
(B) $f_5 \left(\frac{\pi}{28} \right)$	(Q) $\sqrt{3+2\sqrt{2}}$
(C) $f_7 \left(\frac{\pi}{60} \right)$	(R) $\sqrt{2} - 1$
	(S) $\sqrt{7+4\sqrt{3}}$

ASSERTION-REASONING QUESTIONS

This section contains two questions. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which ONLY ONE is correct.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False

(D) Statement -1 is False, Statement-2 is True.

27. Let α, β and γ satisfy $0 < \alpha < \beta < \gamma < 2\pi$ and

$$\cos(x + \alpha) + \cos(x + \beta) + \cos(x + \gamma) = 0 \quad \forall x \in R$$

Statement 1 : $\gamma - \alpha = \frac{2\pi}{3}$

Statement 2: $\cos \alpha + \cos \beta + \cos \gamma = 0$ and $\sin \alpha + \sin \beta + \sin \gamma = 0$

28. Statement 1 : In any triangle ABC

$$\ln \left(\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} \right)$$

$$= \ln \cot \frac{A}{2} + \ln \cot \frac{B}{2} + \ln \cot \frac{C}{2}$$

Statement 2: $\ln(1 + \sqrt{3} + (2 + \sqrt{3}))$

$$= \ln 1 + \ln \sqrt{3} + \ln(2 + \sqrt{3})$$

29. Let $0 < a, b, g < \frac{\pi}{2}$

Statement-1: If $\tan^3 a, \tan^3 b, \tan^3 g$ are the roots of the cubic equation $x^3 - 6x^2 + kx - 8 = 0$, then $\tan a = \tan b = \tan g$.

because

Statement-2: If $a^3 + b^3 + c^3 = 3abc$ and a, b, c are positive numbers then $a = b = c$.

INTEGER QUESTIONS

30. If $x \in [0, 2\pi]$ for which

$$2 \cos x \leq \left| \sqrt{1 + \sin 2x} - \sqrt{1 - \sin 2x} \right| \leq \sqrt{2}$$

has solution set $x \in \left[\frac{\lambda\pi}{4}, \frac{\mu\pi}{4} \right]$ then $\mu - \lambda$ is

31. The number of integral values of α such that $\sin x \cos 3x - \alpha \cos x \sin 3x = 0$ does not have any real root other than $(2n+1)\frac{\pi}{2}, n \in 1$ for any real value of x , is

32. If $4 \cos 36^\circ + \cot \left(7 \frac{1^\circ}{2} \right) =$

$$\sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3} + \sqrt{n_4} + \sqrt{n_5} + \sqrt{n_6}$$

then the product of the digits in $\sum_{i=1}^6 n_i^2 =$

33. If $\tan^2 \left(\frac{\pi}{16} \right) + \tan^2 \left(\frac{2\pi}{16} \right) + \tan^2 \left(\frac{3\pi}{16} \right)$

$$+ \dots + \tan^2 \left(\frac{7\pi}{16} \right) = \lambda \quad \text{and if}$$

$x^y + y^x = \lambda (x, y \in Z^+)$ then the sum of the digits in $(x + y)$ is

34. Suppose $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$ and

$$\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7} = -\frac{1}{8}, \text{ then the numerical}$$

value of $\operatorname{cosec}^2 \frac{\pi}{7} + \operatorname{cosec}^2 \frac{2\pi}{7} + \operatorname{cosec}^2 \frac{3\pi}{7}$
must be

SUBJECTIVE TYPE

35. If $0 < x < y < \frac{\pi}{2}$, show that
 $x - \sin x < y - \sin y$
36. Let a, b, c, d be real numbers such that
 $a + b + c + d = 10$, if the minimum value of
 $a^2 \cot 9^\circ + b^2 \cot 27^\circ + c^2 \cot 63^\circ + d^2 \cot 81^\circ$
 is $\sqrt{n}$ ($n \in \mathbb{N}$) then find n
37. If
 $\sin x + \sin y + \sin z = 0 = \cos x + \cos y + \cos z$
 then show that
 $\cos(\theta - x) + \cos(\theta - y) + \cos(\theta - z) \quad \forall \theta \in \mathbb{R}$
 is independent of x, y, z and θ .
38. If $\frac{\cos \alpha}{\cos A} + \frac{\sin \alpha}{\sin A} = \frac{\cos \beta}{\cos A} + \frac{\sin \beta}{\sin A} = 1$,
 where α and β do not differ by an even
 multiple of π , prove that
 $\frac{\cos \alpha \cos \beta}{\cos^2 A} + \frac{\sin \alpha \sin \beta}{\sin^2 A} = -1$
39. A right angle is divided into three positive
 parts α, β and γ . Prove that for all
 possible divisions $\tan \alpha + \tan \beta + \tan \gamma$
 $> 1 + \tan \alpha \tan \beta \tan \gamma$

LEVEL - VI - KEY**SINGLE ANSWER QUESTIONS**

- 1) B 2) C 3) B 4) A
 5) D 6) D 7) A 8) C
 9) A 10) A 11) B 12) B
 13) C

MULTIPLE ANSWER QUESTIONS

- 14) A, B, C, D 15) A, B, D
 16) A, B 17) C, D
 18) B, D 19) A, B, C, D

COMPREHENSION QUESTIONS

- 20) A 21) A, C, D 22) C
 23) A 24) D 25) B

MATRIX MATCH QUESTIONS

26. (A) R; (B) Q; (C) S

ASSERTION-REASONING QUESTIONS

- 27) D 28) B 29) A
INTEGER TYPE QUESTIONS
 30) 6 31) 3 32) 9 33) 8
 34) 8

SUBJECTIVE QUESTIONS

- 36) 125

LEVEL-VI - HINTS**SINGLE ANSWER QUESTIONS**

1. $\cos A \cos(B+C) - \sin A \sin(B+C)$
 $= \cos A \cos B \cos C$
 $\cos A (\cos(B+C) - \cos B \cos C)$
 $= \sin A \sin(B+C)$
 $\sin(B+C) = -\frac{\cos A \sin B \sin C}{\sin A}$
 $\sin(C+A) = \frac{-\cos B \sin C \sin A}{\sin B}$;
 $\sin(A+B) = \frac{-\cos C \sin A \sin B}{\sin C}$
2. $\sin^4 \frac{\pi}{16} + \cos^4 \frac{\pi}{16} + \sin^4 \frac{3\pi}{16} + \cos^4 \frac{3\pi}{16}$
 $= 1 - \frac{1}{2} \sin^2 \frac{\pi}{8} + 1 - \frac{1}{2} \sin^2 \frac{3\pi}{8}$
 $= 2 - \frac{1}{2} \left(\sin^2 \frac{\pi}{8} + \cos^2 \frac{\pi}{8} \right) = \frac{3}{2}$
3. $(b + a \cos \theta)^2 = 1 - a^2 \sin^2 \theta$

$$(d + c \cos \theta)^2 = 1 - c^2 \sin^2 \theta$$

$$(b + a \cos \theta)^2 (d + c \cos \theta)^2 = (-ac \sin^2 \theta)^2$$

$$(1 - a^2 \sin^2 \theta)(1 - c^2 \sin^2 \theta) = a^2 c^2 \sin^4 \theta$$

$$\Rightarrow (a^2 + c^2) \sin^2 \theta = 1$$

4. $a \sin x + b \cos(x + \theta) + b \cos(x - \theta) = d \Rightarrow a \sin x + 2b \cos x \cos \theta = d$

$$\Rightarrow |d| \leq \sqrt{a^2 + 4b^2 \cos^2 \theta}$$

$$\Rightarrow \frac{d^2 - a^2}{4b^2} \leq \cos^2 \theta$$

$$\Rightarrow |\cos \theta| \geq \frac{\sqrt{d^2 - a^2}}{2|b|}$$

5. $\left(\cos^6 \frac{\pi}{16} + \sin^6 \frac{\pi}{16} \right) + \left(\cos^6 \frac{3\pi}{16} + \sin^6 \frac{3\pi}{16} \right) =$
 $= 1 - 3 \sin^2 \frac{\pi}{16} \cos^2 \frac{\pi}{16} + 1 - 3 \sin^2 \frac{3\pi}{16} \cos^2 \frac{3\pi}{16} = \frac{5}{4}$

6. $\cot^4 x - 2(1 + \cot^2 x) + a^2 = 0$
 $\Rightarrow \cot^4 x - 2 \cot^2 x + a^2 - 2 = 0$
 $\Rightarrow (\cot^2 x - 1)^2 = 3 - a^2$

To have at least one solution, $3 - a^2 \geq 0$

$$\Rightarrow a^2 - 3 \leq 0$$

$$a \in [-\sqrt{3}, \sqrt{3}]$$

Integral values are -1, 0, 1; therefore, the sum is 0.

7. $\sum_{r=1}^n \frac{\tan\left(\frac{\theta}{2^r}\right)}{2^{r-1} \cos\left(\frac{\theta}{2^{r-1}}\right)}$

$$\sum_{r=1}^n \frac{\sin\left(\frac{\theta}{2^r}\right)}{2^{r-1} \cos\left(\frac{\theta}{2^r}\right) \cos\left(\frac{\theta}{2^{r-1}}\right)}$$

$$\sum_{r=1}^n \frac{2 \sin^2\left(\frac{\theta}{2^r}\right)}{2^{r-1} \left\{ 2 \sin\left(\frac{\theta}{2^r}\right) \cos\left(\frac{\theta}{2^r}\right) \right\} \cos\left(\frac{\theta}{2^{r-1}}\right)}$$

$$\sum_{r=1}^n \frac{1 - \cos\left(\frac{\theta}{2^{r-1}}\right)}{2^{r-1} \sin\left(\frac{\theta}{2^{r-1}}\right) \cos\left(\frac{\theta}{2^{r-1}}\right)}$$

$$\sum_{r=1}^n \frac{1}{2^{r-2} \sin\left(\frac{\theta}{2^{r-2}}\right)} - \frac{1}{2^{r-1} \sin\left(\frac{\theta}{2^{r-1}}\right)}$$

$$= \lim_{n \rightarrow \infty} \left[\frac{2}{\sin 2\theta} - \frac{1}{2^{n-1} \sin\left(\frac{\theta}{2^{n-1}}\right)} \right]$$

8. $\sin \theta \cdot \sec 3\theta + \sin 3\theta \cdot \sec 3^2 \theta$
 $+ \sin 3^2 \theta \cdot \sec 3^3 \theta + \dots \dots \dots$ upto n terms

$$= \sum_{r=1}^n \sin 3^{r-1} \theta \cdot \sec 3^r \theta$$

$$= \sum_{r=1}^n \frac{2 \cos 3^{r-1} \theta \sin 3^{r-1} \theta}{2 \cos 3^{r-1} \theta \cdot \cos 3^r \theta}$$

$$= \frac{1}{2} \sum_{r=1}^n \frac{\sin(2 \cdot 3^{r-1} \theta)}{\cos 3^{r-1} \theta \cdot \cos 3^r \theta}$$

$$= \frac{1}{2} \sum_{r=1}^n \frac{\sin(3^r \theta - 3^{r-1} \theta)}{\cos 3^{r-1} \theta \cdot \cos 3^r \theta}$$

$$= \frac{1}{2} \sum_{r=1}^n \frac{\sin 3^r \theta \cdot \cos 3^{r-1} \theta - \cos 3^r \theta \cdot \sin 3^{r-1} \theta}{\cos 3^{r-1} \theta \cdot \cos 3^r \theta}$$

$$\frac{1}{2} \sum_{r=1}^n (\tan 3^r \theta - \tan 3^{r-1} \theta)$$

9. $\frac{\sec^8 \theta}{a} + \frac{\tan^8 \theta}{b} = \frac{1}{a+b}$

$$\text{or } a^2 \sin^8 \theta + ab + b^2 = ab(\cos^8 \theta - \sin^8 \theta)$$

$$= ab(1 - 2 \sin^2 \theta \cos^2 \theta) \cos 2\theta$$

$$= ab(1 - 2 \sin^2 \theta) \left(1 - \frac{1}{2} \sin^2 2\theta \right)$$

$$\Rightarrow a^2 \sin^8 \theta - 2ab \sin^4 \theta + b^2 = -2ab \sin^2 \theta$$

$$+ ab \sin^2 \theta \sin^2 2\theta - \frac{ab}{2} \sin^2 2\theta - 2ab \sin^4 \theta$$

$$(a \sin^4 \theta - b)^2 = 2ab \sin^2 \theta$$

$$\begin{aligned}
 & (-\sin^2 \theta - 1 + 2\sin^2 \theta \cos^2 \theta - \cos^2 \theta) \\
 & \Rightarrow 2ab \sin^2 \theta (-2 + 2\sin^2 \theta \cos^2 \theta) \geq 0 \\
 & \Rightarrow 4ab \sin^2 \theta (\sin^4 \theta - \sin^2 \theta + 1) \leq 0 \\
 & \Rightarrow ab \leq 0
 \end{aligned}$$

10. Given, $f(x) = ax + b$

$$\therefore f'(x) = a$$

since, $a < 0$ $f(x)$ is a decreasing function

$$\therefore f(-1) = 2 \text{ and } f(1) = 0$$

$$\Rightarrow -a + b = 2 \text{ and } a + b = 0$$

$$\therefore a = -1 \text{ and } b = 1$$

$$\text{Thus, } f(x) = -x + 1$$

$$\text{clearly, } f(0) = 1, f\left(\frac{1}{4}\right) = \frac{3}{4}, f(-2) = 3$$

$$f\left(\frac{1}{3}\right) = \frac{2}{3}, f(-1) = 2$$

$$\text{Also, } A = \frac{1 + \cos 2\theta}{2} + \left(\frac{1 - \cos 2\theta}{2}\right)^2$$

$$= \frac{1}{2} + \frac{1}{2} \cos 2\theta + \frac{1}{4} - \frac{1}{2} \cos 2\theta + \frac{1}{4} \cos^2 2\theta$$

$$= \frac{3}{4} + \frac{1}{4} \left(\frac{1 + \cos 4\theta}{2}\right) = \frac{7}{8} + \frac{1}{8} \cos 4\theta$$

$$\therefore \frac{3}{4} \leq A \leq 1$$

$$\Rightarrow f\left(\frac{1}{4}\right) \leq A \leq f(0)$$

11. Equation first can be written as

$$x \sin a + y \times 2 \sin a \cos a + z \times \sin a (3 - 4 \sin^2 a)$$

$$= 2 \times 2 \sin a \cos a \cos 2a$$

$$\Rightarrow x + 2y \cos a + z(3 + 4 \cos^2 a - 4)$$

$$= 4 \cos a (2 \cos^2 a - 1) \text{ as } \sin a \neq 0$$

$$\Rightarrow 8 \cos^3 a - 4z \cos^2 a - (2y + 4) \cos a + (z - x) = 0$$

$$\Rightarrow \cos^3 a - \left(\frac{z}{2}\right) \cos^2 a - \left(\frac{y+2}{4}\right) \cos a + \left(\frac{z-x}{8}\right) = 0$$

Which shows that $\cos a$ is root of the equation

$$t^3 - \left(\frac{z}{2}\right)t^2 - \left(\frac{y+z}{4}\right)t + \left(\frac{z-x}{8}\right) = 0$$

Similarly, from second and third equation we can verify $\cos b$ and $\cos c$ are the roots of the given equation

12. From the third relation we get

$$\cos \theta \cos \phi + \sin \theta \sin \phi = \sin \beta \sin \gamma$$

$$\Rightarrow \sin^2 \theta \sin^2 \phi = (\cos \theta \cos \phi - \sin \beta \sin \gamma)^2$$

$$\Rightarrow \left(1 - \frac{\sin^2 \beta}{\sin^2 \alpha}\right) \left(1 - \frac{\sin^2 \gamma}{\sin^2 \alpha}\right)$$

$$= \left(\frac{\sin \beta \sin \gamma}{\sin^2 \alpha} - \sin \beta \sin \gamma\right)^2$$

[from the first and second relations]

$$\Rightarrow (\sin^2 \alpha - \sin^2 \beta)(\sin^2 \alpha - \sin^2 \gamma)$$

$$= \sin^2 \beta \sin^2 \gamma (1 - \sin^2 \alpha)^2$$

$$\Rightarrow \sin^4 \alpha (1 - \sin^2 \beta \sin^2 \gamma)$$

$$- \sin^2 \alpha (\sin^2 \beta + \sin^2 \gamma - 2 \sin^2 \beta \sin^2 \gamma) = 0$$

$$\therefore \sin^2 \alpha = \frac{\sin^2 \beta + \sin^2 \gamma - 2 \sin^2 \beta \sin^2 \gamma}{1 - \sin^2 \beta \sin^2 \gamma}$$

and

$$\cos^2 \alpha = \frac{1 - \sin^2 \beta - \sin^2 \gamma + \sin^2 \beta \sin^2 \gamma}{1 - \sin^2 \beta \sin^2 \gamma}$$

$$\Rightarrow \tan^2 \alpha$$

$$= \frac{\sin^2 \beta - \sin^2 \beta \sin^2 \gamma + \sin^2 \gamma - \sin^2 \beta \sin^2 \gamma}{\cos^2 \beta - \sin^2 \gamma (1 - \sin^2 \beta)}$$

$$= \frac{\sin^2 \beta \cos^2 \gamma + \cos^2 \beta \sin^2 \gamma}{\cos^2 \beta \cos^2 \gamma}$$

$$= \tan^2 \beta + \tan^2 \gamma$$

$$\Rightarrow \tan^2 \alpha - \tan^2 \beta - \tan^2 \gamma = 0$$

13. Let $S = \sum_{k=1}^{100} \sin(kx) \cos(101-k)x$

$$P S = \sin x \cos 100x + \sin 2x \cos 99x + \dots + \sin 100x \cos x \quad \dots (i)$$

$$S = \cos x \sin 100x + \cos 2x \sin 99x + \dots + \sin x \cos 100x \quad \dots (ii)$$

(on writing in reverse order)

Adding, we get

$$2S =$$

$$\underbrace{\sin 101x + \sin 101x + \dots + \sin 101x}_{100 \text{ times}}$$

$$\text{Hence } S = \frac{100}{2} \sin(101x) = 50 \sin(101x) \text{ Ans.]}$$

MULTIPLE ANSWER QUESTIONS

$$14. \quad x = \sqrt{a^2 \cos^2 \alpha + b^2 \sin^2 \alpha}$$

$$+ \sqrt{a^2 \sin^2 \alpha + b^2 \cos^2 \alpha}$$

$$\Rightarrow x^2 = a^2 + b^2 + 2 \sqrt{(a^2 \cos^2 \alpha + b^2 \sin^2 \alpha)(a^2 \sin^2 \alpha + b^2 \cos^2 \alpha)}$$

$$= a^2 + b^2 + k, \text{ where}$$

$$k = \sqrt{[(a^2 + b^2) - (a^2 \sin^2 \alpha + b^2 \cos^2 \alpha)] \times a^2 \sin^2 \alpha + b^2 \cos^2 \alpha}$$

$$\therefore x = a^2 + b^2 + 2\sqrt{(a^2 + b^2)P - P^2}$$

$$\text{where } P = a^2 \sin^2 \alpha + b^2 \cos^2 \alpha$$

$$\text{or } P = \frac{a^2}{2}(1 - \cos 2\alpha) + \frac{b^2}{2}(1 + \cos 2\alpha)$$

$$= \frac{1}{2}[a^2 + b^2 - (a^2 - b^2)\cos 2\alpha]$$

also,

$$x^2 = \frac{a^2 + b^2 + 2\sqrt{(a^2 \cos^2 \alpha + b^2 \sin^2 \alpha)(a^2 \sin^2 \alpha + b^2 \cos^2 \alpha)}}{[(a^2 + b^2) - (a^2 \cos^2 \alpha + b^2 \sin^2 \alpha)]}$$

$$a^2 + b^2 + 2\sqrt{(a^2 + b^2)P - P^2}$$

$$\text{where, } P = a^2 \cos^2 \alpha + b^2 \sin^2 \alpha$$

$$\text{or } P = \frac{1}{2}[a^2 + b^2 + (a^2 - b^2)\cos 2\alpha]$$

$$15. \quad \text{Let } \tan(\theta/2) = \alpha \text{ and } \tan(\phi/2) = \beta, \text{ so that } \alpha - \beta = 2b$$

$$\text{Also } \cos \theta = \frac{1 - \tan^2(\theta/2)}{1 + \tan^2(\theta/2)} = \frac{1 - \alpha^2}{1 + \alpha^2}$$

$$\text{and } \sin \theta = \frac{2 \tan(\theta/2)}{1 + \tan^2(\theta/2)} = \frac{2\alpha}{1 + \alpha^2}$$

$$\text{similarly } \cos \phi = \frac{1 - \beta^2}{1 + \beta^2} \text{ and } \sin \phi = \frac{2\beta}{1 + \beta^2}$$

Therefore, we have from the given relations

$$(x - a) \frac{1 - \alpha^2}{1 + \alpha^2} + y \left(\frac{2\alpha}{1 + \alpha^2} \right) = a$$

$$\Rightarrow x\alpha^2 - 2y\alpha + 2a - x = 0$$

$$\text{similarly } x\beta^2 - 2y\beta + 2a - x = 0$$

We see that α and β are the roots of the equation $xz^2 - 2yz + 2a - x = 0$, so that $\alpha + \beta = 2y/x$ and $\alpha\beta = (2a - x)/x$. Now, from $(\alpha + \beta)^2 = (\alpha - \beta)^2 + 4\alpha\beta$, we get

$$\left(\frac{2y}{x} \right)^2 = (2b)^2 + \frac{4(2a - x)}{x}$$

$$\Rightarrow y^2 = 2ax - (1 - b^2)x^2$$

Also, from $\alpha + \beta = 2y/x$ and $\alpha - \beta = 2b$, we get

$$\alpha = y/x + b \text{ and } \beta = y/x - b$$

$$\Rightarrow \tan \frac{\theta}{2} = \frac{1}{x}(y + bx) \text{ and } \tan \frac{\phi}{2} = \frac{1}{x}(y - bx)$$

16. The given equation can be written as

$$\frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} - \sin \alpha \cot \beta \frac{2 \tan(x/2)}{1 + \tan^2(x/2)} = \cos \alpha$$

$$\Rightarrow \tan^2 \frac{x}{2} (1 + \cos \alpha) + \sin \alpha \cot \beta \cdot 2 \tan \frac{x}{2}$$

$$- (1 - \cos \alpha) = 0$$

$$\Rightarrow \tan^2 \frac{x}{2} + \frac{2 \sin \alpha \cot \beta}{1 + \cos \alpha} \tan \frac{x}{2} - \frac{1 - \cos \alpha}{1 + \cos \alpha} = 0$$

JEE MAINS ADVANCED

$$\Rightarrow \tan^2 \frac{x}{2} + 2 \tan \frac{\alpha}{2} \cot \beta \tan \frac{x}{2} - \tan^2 \frac{\alpha}{2} = 0$$

$$\Rightarrow \tan^2 \frac{x}{2} + 2 \tan \frac{\alpha}{2} \cdot \frac{1}{2} \left(\cot \frac{\beta}{2} - \tan \frac{\beta}{2} \right) \tan^2 \frac{\alpha}{2} = 0$$

$$\Rightarrow \left(\tan \frac{x}{2} + \cot \frac{\beta}{2} \tan \frac{\alpha}{2} \right) \left(\tan \frac{x}{2} - \tan \frac{\beta}{2} \tan \frac{\alpha}{2} \right) = 0$$

$$\Rightarrow \tan(x/2) = -\tan(\alpha/2) \cot(\beta/2)$$

$$\text{or } \tan(x/2) = \tan(\alpha/2) \tan(\beta/2)$$

17. $(\sin x + \cos x)$

$$+ \left(\frac{1}{\sin x \cos x} \right) + \left(\frac{\sin x + \cos x}{\sin x \cos x} \right) = 7$$

$$\Rightarrow (\sin x + \cos x) \left(1 + \frac{1}{\sin x \cos x} \right) = 7 - \frac{2}{\sin 2x}$$

$$\text{squaring both sides } \Rightarrow \sin^3 2x - 44 \sin^2 2x + 36 \sin 2x = 0$$

$$\Rightarrow \sin^2 2x - 44 \sin 2x + 36 = 0 \quad (\sin 2x \neq 0)$$

$$\Rightarrow \sin 2x = 22 - 8\sqrt{7}$$

18. Expression = $(a \tan g + b \cot g)(a \cot g + b \tan g) - 4ab \cot^2 2g$

$$= a^2 + (\tan^2 g + \cot^2 g)ab + b^2 - 4ab \frac{\cos^2 2g}{\sin^2 2g}$$

$$= a^2 + b^2 + ab$$

$$\left[\left(\frac{\sin^2 \gamma}{\cos^2 \gamma} + \frac{\cos^2 \gamma}{\sin^2 \gamma} \right) - \frac{4(\cos^2 \gamma - \sin^2 \gamma)^2}{4 \sin^2 \gamma \cos^2 \gamma} \right]$$

$$= a^2 + b^2 + ab$$

$$\left[\frac{(\sin^4 \gamma + \cos^4 \gamma) - (\cos^4 \gamma + \sin^4 \gamma - 2 \sin^2 \gamma \cos^2 \gamma)}{\sin^2 \gamma \cos^2 \gamma} \right]$$

$$= a^2 + b^2 + 2ab = (a + b)^2,$$

which is independent of g and dependent on a, b .]

19. Let $x = \cos q$

[12th, 13th 07-01-2007]

$$4 \cos^3 q - 3 \cos q = -\frac{\sqrt{3}}{2}$$

$$\cos 3q = \cos \frac{5\pi}{6} \quad \text{P } 3q = 2n\pi \pm$$

$$\frac{5\pi}{6} \quad \text{P } q = \frac{2n\pi}{3} \pm \frac{5\pi}{18}$$

$$\text{put } n = 0, \quad q = \frac{5\pi}{18}$$

$$n = 1, \quad q = \frac{2\pi}{3} + \frac{5\pi}{18} = \frac{17\pi}{18};$$

$$q = \frac{2\pi}{3} - \frac{5\pi}{18} = \frac{7\pi}{18}$$

COMPREHENSION QUESTIONS

Passage: 20,21,22

$$0 < \alpha < 90^\circ, 90^\circ < \beta < 180^\circ, 180^\circ$$

$$< \gamma < 270^\circ, 270^\circ < \delta < 360^\circ$$

$$\Rightarrow 270^\circ < \alpha + \delta < 450^\circ \Rightarrow \alpha + \delta$$

lies in the I or IV

quadrant and cosine in both is positive

If d is the common ratio of the A.P

$$\text{Then } \beta = \alpha + d, \gamma = \alpha + 2d, \delta = \alpha + 3d$$

$$\Rightarrow \beta + \gamma = \alpha + \delta, 2(\alpha - \beta) = -2d = \beta - \delta$$

and $\alpha + \gamma = 2\beta$. Note

$$\beta - \gamma = -d, \alpha - \delta = -3d \quad 270^\circ < \delta < 360^\circ$$

$$\Rightarrow 270^\circ < \alpha + 3d < 360^\circ$$

$$\Rightarrow 200^\circ < 3d < 290^\circ$$

$$[\therefore \alpha = 70^\circ]$$

$$\Rightarrow 400^\circ < 6d < 580^\circ$$

$$\Rightarrow 680^\circ < 4\alpha + 6d < 860^\circ$$

$$\Rightarrow 680^\circ < \theta < 860^\circ$$

Passage: 23,24,25

$$OL = r \cos(\alpha - 3\theta)$$

$$CL = r \sin(\alpha - 3\theta)$$

$$\text{so } \frac{\sin^3 \theta}{r \sin(\alpha - 3\theta)} = \frac{\cos^3 \theta}{r \cos(\alpha - 3\theta)} = 1$$

$$\Rightarrow \frac{\sin^3 \theta}{\sin(\alpha - 3\theta)} = \frac{\cos^3 \theta}{\cos(\alpha - 3\theta)} = r$$

$$\text{Now } \cos^4 \theta - \sin^4 \theta = \cos \theta (r \cos(\alpha - 3\theta)) - \sin \theta (r \sin(\alpha - 3\theta))$$

$$\Rightarrow \cos 2\theta = r \cos(\alpha - 2\theta)$$

$$\Rightarrow \cos 2\theta = r(\cos \alpha \cos 2\theta + \sin \alpha \sin 2\theta)$$

$$\Rightarrow (1 - r \cos \alpha) \cos 2\theta = r \sin \alpha \sin 2\theta$$

$$\Rightarrow \frac{1 - r \cos \alpha}{r \sin \alpha} = \tan 2\theta$$

$$\text{Next, } \cos^3 \theta \sin \theta + \sin^3 \theta \cos \theta$$

$$= r[\cos(\alpha - 3\theta) \sin \theta + \sin(\alpha - 3\theta) \cos \theta]$$

$$\Rightarrow \sin \theta \cos \theta = r \sin(\alpha - 2\theta)$$

$$\Rightarrow \sin 2\theta = 2r[\sin \alpha \cos 2\theta - \cos \alpha \sin 2\theta]$$

$$\Rightarrow (1 + 2r \cos \alpha) \sin 2\theta = 2r \sin \alpha \cos 2\theta$$

$$\Rightarrow \frac{1 + 2r \cos \alpha}{2r \sin \alpha} = \cot 2\theta$$

from (A) and (B) we get

$$\frac{1 - r \cos \alpha}{r \sin \alpha} = \frac{2r \sin \alpha}{1 + 2r \cos \alpha}$$

$$\Rightarrow 1 - r \cos \alpha + 2r \cos \alpha - 2r^2 \cos^2 \alpha = 2r^2 \sin^2 \alpha$$

$$\Rightarrow \frac{2r^2 - 1}{r} = \cos \alpha$$

MATRIX MATCHING QUESTIONS

26. We have $f_n(q) = \cot\left((3n-1)\frac{\theta}{4}\right)$

$$(A) f_3\left(\frac{3\pi}{16}\right) = \cot\left(8 \times \frac{1}{4} \times \frac{3\pi}{16}\right) = \cot \frac{3\pi}{8}$$

$$= \sqrt{2} - 1$$

$$(B) f_5\left(\frac{\pi}{28}\right) = \cot\left(14 \times \frac{1}{4} \times \frac{\pi}{28}\right)$$

$$= \cot \frac{\pi}{8} = \sqrt{2} + 1 = \sqrt{3 + 2\sqrt{2}}$$

$$(C) f_7\left(\frac{\pi}{60}\right) = \cot\left(20 \times \frac{1}{4} \times \frac{\pi}{60}\right) = \cot \frac{\pi}{12}$$

$$= (2 + \sqrt{3}) = \sqrt{7 + 4\sqrt{3}} \quad]$$

ASSERTION-REASONING QUESTIONS

27. Given

$$\cos x \sum \cos \alpha - \sin x \sum \sin \alpha = 0 \forall x \in R$$

$$\text{hence } \cos \alpha + \cos \beta + \cos \gamma = 0$$

$$\text{and } \sin \alpha + \sin \beta + \sin \gamma = 0$$

$$\Rightarrow \text{statement 2 is true}$$

$$\text{Now } (\cos \alpha + \cos \beta)^2 = (-\cos \gamma)^2$$

$$\text{and } (\sin \alpha + \sin \beta)^2 = (-\sin \gamma)^2$$

Adding, we get

$$2 + 2 \cos(\alpha - \beta) = 1$$

$$\Rightarrow \cos(\alpha - \beta) = -1/2$$

$$\text{similarly } \cos(\beta - \gamma) = -1/2 \text{ and}$$

$$\cos(\gamma - \alpha) = -1/2$$

$$\text{Now } 0 < \alpha < \beta < \gamma < 2\pi$$

$$\Rightarrow \beta - \alpha < \gamma - \alpha$$

$$\text{Hence, } \beta - \alpha = \frac{2\pi}{3} \text{ and } \gamma - \alpha = \frac{4\pi}{3} \Rightarrow$$

statement 1 is false

28. In triangle ABC, $A+B+C = \pi$
and

$$\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$$

$$\Rightarrow \ln\left(\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}\right)$$

$$= \ln \cot \frac{A}{2} + \ln \cot \frac{B}{2} + \ln \cot \frac{C}{2}$$

Hence, statement 1 is true. In statement 2, R.H.S

$$= \ln 1 + \ln \sqrt{3} + \ln(2 + \sqrt{3})$$

$$= \ln(1\sqrt{3}(2+\sqrt{3})) = \ln(2\sqrt{3}+3) = R.H.S$$

But statement 2 does not explain statement 1.

29. We have $\tan^3 a + \tan^3 b + \tan^3 g = 6$

$$\tan^3 a \tan^3 b \tan^3 g = 8$$

$$\text{P } \tan a \tan b \tan g = 2$$

Hence $\tan^3 a + \tan^3 b + \tan^3 g = 3 \tan a \tan b \tan g$

$$\text{P } \frac{1}{2}(\tan \alpha + \tan \beta + \tan \gamma)$$

$$[(\tan \alpha - \tan \beta)^2 + (\tan \beta - \tan \gamma)^2 + (\tan \gamma - \tan \alpha)^2] = 0$$

But $\tan a + \tan b + \tan g > 0$

$\text{P } \tan a = \tan b = \tan g$]

INTEGER QUESTIONS

30. Let $y = |\sqrt{1+\sin 2x} - \sqrt{1-\sin 2x}|$

$$\Rightarrow y^2 = 2 - 2|\cos 2x|$$

$$\text{If } x \in \left[0, \frac{\pi}{4}\right] \text{ or } \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right] \text{ or } \left[\frac{7\pi}{4}, 2\pi\right]$$

$\cos 2x$ is non - negative

$$\text{so, } y^2 = 2 - 2\cos 2x = 4\sin^2 x$$

$$y = 2|\sin x| \Rightarrow \cos x \leq |\sin x|$$

Except for x in $\left[0, \frac{\pi}{4}\right]$ and $\left[\frac{7\pi}{4}, 2\pi\right]$

so, that leaves $\left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$

In which we certainly have $\sin x \leq \frac{1}{\sqrt{2}}$

If $x \in \left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ or $\left(\frac{5\pi}{4}, \frac{7\pi}{4}\right)$ then

$\cos 2x$ is negative, so

$$y^2 = 2 + 2\cos 2x = 4\cos^2 x$$

$$\Rightarrow y = 2|\cos x|$$

so, the first inequality certainly holds the second also holds

Thus, solution set is $x \in \left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$

$$\Rightarrow \mu - \lambda = 6$$

31. $\sin x \cos 3x - \alpha \cos x \sin 3x = 0$ (i)

$$\Rightarrow \alpha = \frac{\sin x \cos 3x}{\cos x \sin 3x}$$

$$\alpha = \frac{\tan x}{\tan 3x}$$

$$\alpha = \frac{\tan x(1-3\tan^2 x)}{3\tan x - \tan^3 x}$$

$$\alpha = \frac{1-3\tan^2 x}{3-\tan^2 x}$$

....(ii)

For real value of x , RHS of 2nd never lies

between $\left[\frac{1}{3}, 3\right]$

$$\Rightarrow \text{Number of integral points } \{1, 2, 3\}$$

$$\Rightarrow 3 \text{ points}$$

32. $\therefore \cot\left(7\frac{1^\circ}{2}\right) = \frac{1+\cos 15^\circ}{\sin 15^\circ}$

$$= \frac{1+\frac{\sqrt{3}+1}{2\sqrt{2}}}{\frac{\sqrt{3}-1}{2\sqrt{2}}} = \frac{2\sqrt{2}+\sqrt{3}+1}{\sqrt{3}-1}$$

$$= \frac{(2\sqrt{2}+\sqrt{3}+1)(\sqrt{3}+1)}{2}$$

$$\frac{2\sqrt{6}+2\sqrt{2}+3+\sqrt{3}+\sqrt{3}+1}{2}$$

$$= \sqrt{6} + \sqrt{2} + 2 + \sqrt{3}$$

$$= \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{6}$$

and

$$4 \cos 36^\circ = 4 \left(\frac{\sqrt{5}+1}{4} \right) = \sqrt{5} + 1 = \sqrt{5} + \sqrt{1}$$

$$\text{Hence, } 4 \cos 36^\circ + \cot \left(7 \frac{1^\circ}{2} \right)$$

$$= \sqrt{1} + \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{5} + \sqrt{6}$$

$$\therefore n_1 = 1, n_2 = 2, n_3 = 3, n_4 = 4$$

$$n_5 = 5 \text{ and } n_6 = 6$$

$$\therefore \sum_{i=1}^6 n_i^2 = n_1^2 + n_2^2 + n_3^2 + n_4^2 + n_5^2 + n_6^2$$

$$= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2$$

$$= 91$$

$$\begin{aligned} 33. \quad \lambda &= \left\{ \tan^2 \left(\frac{\pi}{16} \right) + \tan^2 \left(\frac{7\pi}{16} \right) \right\} \\ &+ \left\{ \tan^2 \left(\frac{2\pi}{16} \right) + \tan^2 \left(\frac{6\pi}{16} \right) \right\} \\ &+ \left\{ \tan^2 \left(\frac{3\pi}{16} \right) + \tan^2 \left(\frac{5\pi}{16} \right) \right\} + \tan^2 \left(\frac{4\pi}{16} \right) \\ &= \tan^2 \left(\frac{\pi}{16} \right) + \cot^2 \left(\frac{\pi}{2} - \frac{7\pi}{16} \right) \\ &+ \left\{ \tan^2 \left(\frac{2\pi}{16} \right) + \cot^2 \left(\frac{\pi}{2} - \frac{6\pi}{16} \right) \right\} \\ &+ \left\{ \tan^2 \left(\frac{3\pi}{16} \right) + \cot^2 \left(\frac{\pi}{2} - \frac{5\pi}{16} \right) \right\} + 1 \\ &= \left\{ \tan^2 \left(\frac{\pi}{16} \right) + \cot^2 \left(\frac{\pi}{16} \right) \right\} \\ &+ \left\{ \tan^2 \left(\frac{2\pi}{16} \right) + \cot^2 \left(\frac{2\pi}{16} \right) \right\} \\ &+ \left\{ \tan^2 \left(\frac{3\pi}{16} \right) + \cot^2 \left(\frac{3\pi}{16} \right) \right\} + 1 \end{aligned}$$

$$\begin{aligned} &= \left\{ \tan \left(\frac{\pi}{16} \right) + \cot \left(\frac{\pi}{16} \right) \right\}^2 + \left\{ \tan \left(\frac{2\pi}{16} \right) + \cot \left(\frac{2\pi}{16} \right) \right\}^2 \\ &\quad + \left\{ \tan \left(\frac{3\pi}{16} \right) + \cot \left(\frac{3\pi}{16} \right) \right\}^2 - 2 - 2 - 2 + 1 \\ &= \frac{1}{\left\{ \sin \left(\frac{\pi}{16} \right) \cos \left(\frac{\pi}{16} \right) \right\}^2} + \frac{1}{\left\{ \sin \left(\frac{2\pi}{16} \right) \cos \left(\frac{2\pi}{16} \right) \right\}^2} \\ &\quad + \frac{1}{\left\{ \sin \left(\frac{3\pi}{16} \right) \cos \left(\frac{3\pi}{16} \right) \right\}^2} - 5 \\ &= \frac{4}{\sin^2 \frac{\pi}{8}} + \frac{4}{\sin^2 \frac{\pi}{4}} + \frac{4}{\sin^2 \left(\frac{3\pi}{8} \right)} - 5 \\ &= 4 \left\{ \frac{1}{\sin^2 \left(\frac{\pi}{8} \right)} + \frac{1}{\sin^2 \left(\frac{3\pi}{8} \right)} \right\} + 4.2 - 5 \\ &= 4 \left\{ \frac{1}{\sin^2 \left(\frac{\pi}{8} \right)} + \frac{1}{\cos^2 \left(\frac{\pi}{8} \right)} \right\} + 3 \\ &= \frac{4}{\left\{ \sin \left(\frac{\pi}{8} \right) \cos \left(\frac{\pi}{8} \right) \right\}^2} + 3 \\ &= \frac{16}{\left\{ \sin \left(\frac{\pi}{4} \right) \right\}^2} + 3 \end{aligned}$$

$$\therefore \lambda = 35$$

$$\text{Then, } x^y + y^x = 35$$

$$\Rightarrow x = 34, y = 1$$

$$\text{or } x = 1, y = 34$$

$$34. \quad \cos ec^2 \frac{\pi}{7} + \cos ec^2 \frac{2\pi}{7} + \cos ec^2 \frac{3\pi}{7}$$

$$= \frac{1}{\sin^2 \frac{\pi}{7}} + \frac{1}{\sin^2 \frac{2\pi}{7}} + \frac{1}{\sin^2 \frac{3\pi}{7}}$$

$$= \frac{2}{1 - \cos \frac{2\pi}{7}} + \frac{2}{1 - \cos \frac{4\pi}{7}} + \frac{2}{1 - \cos \frac{6\pi}{7}}$$

Now denominator (after taking LCM)

$$= \left(1 - \cos \frac{2\pi}{7}\right) \left(1 - \cos \frac{4\pi}{7}\right) \left(1 - \cos \frac{6\pi}{7}\right)$$

$$= 1 - \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}\right)$$

$$+ \left(\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} + \cos \frac{4\pi}{7} \cos \frac{6\pi}{7} + \cos \frac{6\pi}{7} \cos \frac{2\pi}{7}\right)$$

$$- \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7}$$

The second and last terms are known in the light of given relations.

The third term

$$= \frac{1}{2} \left[\cos \frac{6\pi}{7} + \cos \frac{2\pi}{7} + \cos \frac{10\pi}{7} + \cos \frac{2\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{4\pi}{7} \right]$$

$$= \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$$

$$\text{This denominator} = 1 - \left(-\frac{1}{2}\right) - \frac{1}{2} + \frac{1}{8} = \frac{7}{8}$$

The numerator

$$= 2 \left[3 - 2 \left[\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right] \right]$$

$$+ \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} + \cos \frac{4\pi}{7} \cos \frac{6\pi}{7} + \cos \frac{6\pi}{7} \cos \frac{2\pi}{7}$$

$$= 2 \left[3 - 2 \left(-\frac{1}{2} \right) - \frac{1}{2} \right] = 7. \text{ Whence the given}$$

$$\text{expression} = \frac{7}{7/8} = 8$$

SUBJECTIVE QUESTIONS

$$35. \sin y - \sin x = 2 \sin \frac{y-x}{2} \cos \frac{y+x}{2} < 2 \sin \frac{y-x}{2}$$

$$\therefore \sin y - \sin x < 2 \sin \frac{y-x}{2} \sin \theta < \theta$$

$$\therefore x - \sin x < y - \sin y$$

36. If $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are $2n$ real numbers, then

$$(a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2$$

$$\leq (a_1^2 + a_2^2 + \dots + a_n^2) (b_1^2 + b_2^2 + \dots + b_n^2)$$

.... (1)

$$\text{Let } a_1 = a \sqrt{\cot 9^\circ}, \quad a_2 = b$$

$$\sqrt{\cot 27^\circ}, \quad a_3 = c \sqrt{\cot 63^\circ}, \quad a_4 = d \sqrt{\cot 81^\circ}$$

$$\text{and } b_1 = \sqrt{\tan 9^\circ}, \quad b_2 = \sqrt{\tan 27^\circ}, \quad b_3$$

$$= \sqrt{\tan 63^\circ}, \quad b_4 = \sqrt{\tan 81^\circ}$$

Now using (1), we get

$$(a + b + c + d)^2 \leq (a^2 \cot 9^\circ + b^2 \cot 27^\circ + c^2 \cot 63^\circ + d^2 \cot 81^\circ)$$

$$(\tan 9^\circ + \tan 27^\circ + \tan 63^\circ + \tan 81^\circ) \dots (2)$$

But $a + b + c + d = 10$ (Given)

$$\text{and } (\tan 9^\circ + \tan 81^\circ) + (\tan 27^\circ + \tan 63^\circ)$$

$$= \frac{1}{\sin 9^\circ \cos 9^\circ} + \frac{1}{\sin 27^\circ \cos 27^\circ} = \frac{2}{\sin 18^\circ}$$

$$+ \frac{2}{\sin 54^\circ} = \frac{2}{\sqrt{5}-1} + \frac{2}{\sqrt{5}+1}$$

$$\frac{4}{4} = 1$$

$$= 8 \left[\frac{(\sqrt{5}+1) + (\sqrt{5}-1)}{4} \right] = 4\sqrt{5}$$

\ From (2), we get

$$100 \leq 4\sqrt{5} (a^2 \cot 9^\circ + b^2 \cot 27^\circ + c^2 \cot 63^\circ + d^2 \cot 81^\circ)$$

$$\text{P } a^2 \cot 9^\circ + b^2 \cot 27^\circ + c^2 \cot 63^\circ + d^2 \cot$$

$$81^{\circ 3} \frac{25}{\sqrt{5}} = 5\sqrt{5} = \sqrt{125} = \sqrt{n}$$

Hence $n = 125$]

37. Conceptual

38. α and β are the roots of the equation

$$\frac{\cos \theta}{\cos A} + \frac{\sin \theta}{\sin A} = 1$$

$$\text{or, } \left(\frac{\cos \theta}{\cos A} \right)^2 = \left(1 - \frac{\sin \theta}{\sin A} \right)^2$$

On simplification

$$\left(\frac{\sin \theta}{\sin A} \right)^2 - 2 \left(\frac{\sin \theta}{\sin A} \right) \cos^2 A - \sin^2 A = 0$$

Clearly,

$\frac{\sin \alpha}{\sin A}$ and $\frac{\sin \beta}{\sin A}$ are the roots of this equation

$$\therefore \frac{\sin \alpha}{\sin A} \frac{\sin \beta}{\sin A} = -\sin^2 A$$

$$\Rightarrow \frac{\sin \alpha \sin \beta}{\sin^2 A} = -\sin^2 A$$

Similarly, by making a quadratic in $\frac{\cos \theta}{\cos A}$, we obtain

$$\frac{\cos \alpha}{\cos A} \cdot \frac{\cos \beta}{\cos A} = -\cos^2 A$$

$$\Rightarrow \frac{\cos \alpha \cos \beta}{\cos^2 A} = -\cos^2 A$$

Adding the above equations

$$\begin{aligned} & \frac{\cos \alpha \cos \beta}{\cos^2 A} + \frac{\sin \alpha \sin \beta}{\sin^2 A} \\ &= -(\cos^2 A + \sin^2 A) = -1 \end{aligned}$$

$$\begin{aligned} 39. \quad y &= \tan \alpha + \tan \beta + \tan \gamma \\ &\quad - \tan \alpha \tan \beta \tan \gamma \end{aligned}$$

$$\Rightarrow y = \frac{\sin(\alpha + \beta + \gamma)}{\cos \alpha \cos \beta \cos \gamma}$$

$$\Rightarrow y = \frac{1}{\cos \alpha \cos \beta \cos \gamma} \left[\because \alpha + \beta + \gamma = \frac{\pi}{2} \right]$$

$$\Rightarrow y = \frac{2}{(\cos(\beta + \gamma) + \cos(\beta - \gamma)) \cos \alpha}$$

$$\Rightarrow y = \frac{2}{(\sin \alpha + \cos(\beta - \gamma)) \cos \alpha}$$

For given value of α and varying β and γ , we find that y is minimum if $\cos(\beta - \gamma)$ is maximum i.e. $\beta = \gamma$

Similarly, for given β , y is minimum if $\alpha = \gamma$

Thus, y is minimum, if

$$\alpha = \beta = \gamma = \frac{\pi}{6}$$